



REGIONAL CAPACITY BUILDING TRAINING WORKSHOP ON CLIMATE BASICS AND CLIMATE INFORMATION FOR AGRICULTURAL EXPERTS IN SOUTHERN AND EASTERN AFRICA REGIONS



Training Report

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About AICCRA Reports

The titles in this series aim to disseminate interim research on scaling climate services and climate-smart agriculture in Africa and stimulate feedback from the scientific community.

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ABOUT AICCRA



AICCRA
Accelerating Impacts of CGIAR
Climate Research for Africa



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ABSTRACT

Climate change presents significant challenges to agriculture in Eastern and Southern Africa, a region heavily reliant on rainfed farming systems. To enhance resilience and productivity, agricultural experts need a strong foundation in climate systems and using Climate Information Services (CIS) effectively. A capacity-building training workshop aimed at equipping agricultural experts in Southern and Eastern Africa with the knowledge and skills to tackle climate change challenges effectively was held in Gaborone, Botswana, from September 16-20, 2024. The training workshop engaged a total of total of 57 participants from 18 countries, including 25% women.

This regional training workshop comprehensively introduced climate basics, climate information services, early warning systems, disaster risk management, and climate change impacts. Additionally, the workshop allowed the participants to access and use different tools, such as East Africa Drought Watch, Agriculture Monitoring for Eastern Africa, and East Africa Hazards Watch.

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ACRONYMS

AICCRA	Accelerating Impacts of CGIAR Climate Research for Africa
ASARECA	Association for Strengthening Agricultural Research in Eastern and Central Africa
CCARDESA	Centre for Coordination of Agricultural Research and Development for Southern Africa
CIS	Climate information services
CSA	Climate-smart agriculture
ICARDA	International Center for Agricultural Research in the Dry Areas
IGAD	Intergovernmental Authority on Development
ILRI	International Livestock Research Institute
IPCC	Intergovernmental Panel on Climate Change



INTRODUCTION

Capacity-building training workshops focusing on climate basics, climate information services, early warning systems, disaster risk management, and climate change impacts are essential for a wide range of Eastern and Southern Africa climate-sensitive sectors due to the region's acute vulnerability to climate variability and change. For instance, the agricultural sector, a cornerstone of these regions' economies, faces significant threats from changing weather patterns and increased frequency and intensity of extreme events (e.g., droughts and floods). According to the Intergovernmental Panel on Climate Change (IPCC), the region is also experiencing warmer temperatures, erratic rainfall and seasonal shifts, and frequent extreme weather events, directly threatening food security and livelihoods (IPCC, 2021). Therefore, agricultural experts should be equipped with knowledge and skills in the basics of climate science and climate information services to better understand these changes and develop strategies for mitigating their impacts on crop and livestock productivity, water availability, and food security.

Moreover, effective disaster risk management and early warning systems are critical to reducing climate-induced adverse impacts. However, many countries in the region lacked the necessary technology, communication and dissemination mechanisms, institutional coordination and plans, and the capacity to build the expertise required to predict and respond to such events in a timely and reliable manner. The South-South knowledge exchange and capacity-building training initiatives organized by regional partners such as the Accelerated Impacts of CGIAR Climate Research for Africa (AICCRA) and IGAD Climate Prediction and Application Center (ICPAC) empower experts in the region to understand and effectively utilize climate information. These initiatives also raise awareness about the four pillars of early warning systems and foster connections with information providers to enhance preparedness and response capabilities. Promoting skills in climate risk management and adaptation strategies, such initiatives help bridge the gap between scientific research and practical application, enabling experts to implement policies and practices that safeguard agricultural productivity and community well-being. The United Nations Framework Convention on Climate Change (UNFCCC) underscores the importance of training initiatives that build adaptive capacity in the regions (UNFCCC, 2018).

Disseminating Climate Information Services (CIS) also involves addressing the gender-sensitive needs of both men and women. To effectively tackle these gender-specific challenges, it is essential to enhance the capacities of male and female agricultural experts and lead farmers. This will enable them to utilize climate information effectively, manage risks, and implement adaptation strategies.



WELCOMING SPEECH



Professor Cliff Sibusiso Dlamini, Executive Director of CCARDESA, welcomed all participants to the regional capacity-building training workshop. He acknowledged representatives from Botswana's agriculture research and development sector, ICPAC, AICCRA, ASARECA, and the FSRP project team, extending heartfelt gratitude to everyone for their presence in Gaborone. He noted that this workshop is critical, providing a valuable opportunity to deepen our understanding of climate science and its connection to agricultural practices.

Professor Dlamini noted that the training will empower participants to understand, interpret, disseminate, and apply climate information more effectively in decision-making processes. He stressed the importance of this knowledge and skill, particularly for climate change adaptation in Africa, where the challenge is not a lack of information but often a lack of capacity to interpret and utilize it efficiently. He underscored the importance of understanding gender dynamics in agriculture to ensure that climate information reaches and benefits all stakeholders equitably. He expressed his appreciation for the collaboration between institutions, including CCARDESA, ASARECA, ICPAC, SADC-CSC, and CGIAR, which are crucial to advancing shared goals in climate adaptation and agricultural resilience across Africa.

In closing, Professor Dlamini expressed confidence that by the end of the workshop, participants will leave with enhanced capacity, better knowledge of climate basics, climate information services, early warning systems, and improved strategies for climate change adaptation. He looked forward to stronger collaboration and knowledge sharing among agriculture stakeholders, fostering resilience in the region's agricultural systems.



Dr. Abebe Tadege, principal investigator at the IGAD Climate Prediction and Application Center (ICPAC), on behalf of Dr. Abdi Fidar, welcomed participants to the training, expressing his deep appreciation to AICCRA, CCARDESA, ASARECA, and SADC-CSC for turning his long-held wish for a workshop like this into reality. Reflecting on his 15-20 years of involvement in climate change negotiations, Dr. Abebe noted that addressing the complex challenges posed by climate change becomes nearly impossible

without a strong foundational understanding. Touching on the workshop's content, Dr. Abebe reiterated Professor Dlamini's emphasis on the significance of topics like understanding climate basics, the role of gender in climate strategies, disaster risk management, distinguishing weather from climate, and recognizing climate variability and



associated risks. He also underscored the importance of climate services, which are increasingly seen as a key mechanism for building resilience to climate change and must be prioritized at global forums like COP.

In closing, he wished everyone a fruitful and productive training workshop and expressed his excitement for the important work ahead.



Mr. Moses Odeke, Interim Head of Programs at ASARECA, began by warmly welcoming distinguished guests and participants. He expressed his honor in joining his colleagues to welcome everyone to this vital regional capacity-building training workshop. He emphasized that while we have long acknowledged that "climate change is real," we now face a rapidly intensifying climate crisis. This crisis has created a vicious cycle of vulnerability, making recovery increasingly difficult, especially for marginalized groups

such as women, children, and the elderly, who bear disproportionate impacts. This workshop, he noted, is a key milestone in our collective efforts to address these urgent challenges and their implications for agriculture.

He urged us to recognize the critical need for climate action. The training, which focuses on topics such as climate basics, EWS, DRM, risk mitigation, and climate adaptation strategies, is designed to provide participants with the essential knowledge, skills, and tools to address these challenges effectively. He added that EWS hold tremendous potential if scaled up. By connecting millions of smallholder farmers to these systems through agro-advisory services, we can help them make informed decisions, thus reducing the impacts of climate change.

In closing, he expressed his sincere gratitude to partners, including ILRI-AICCRA, ICPAC, and CCARDESA, for their excellent collaboration and support in organizing this event. He also thanked the World Bank for supporting the AICCRA project, which made the workshop possible. Lastly, he thanked all participants for their involvement and the expertise they bring to the table.



Dr. Dawit Solomon, Regional and Ethiopia Country Program Leader for Accelerating Impacts of CGIAR Climate Research for Africa (AICCRA), began by addressing the distinguished guests, climate and agricultural experts. He shared a personal reflection, recalling when they first initiated discussions with the CEO of ASARECA, who enthusiastically endorsed the idea of building regional organizations' capacity using Africa's resources. He emphasized the importance of relying on Africa's capacity to tackle its challenges



and how this initiative was enthusiastically supported by ASARECA, CCARDESA, ICPAC, and other organizations. Dr. Dawit also thanked the Food Systems Resilience Project leaders at the World Bank and other key partners for their support.

On behalf of AICCRA's regional program for Eastern and Southern Africa, Dr. Dawit warmly welcomed all attendees at the jointly organized regional climate capacity-building training, which brings together partners from IGAD and SADC member countries. He highlighted the significance of this training in strengthening collective efforts to address the pressing challenges of climate change in Eastern and Southern Africa.

Dr. Dawit sincerely thanked the partners, including IGAD and its FSRP program, CCARDESA, WMO Regional Office for Africa, ACPC, IRI, and SADC CSC. He acknowledged their valuable collaboration and support, which was instrumental in shaping the training's content and curriculum. He further appreciated the collaboration between AICCRA and the World Bank's Food Systems Resilience Project in Eastern and Southern Africa, stating that this partnership has strengthened their ability to address the region's interconnected challenges of climate change and food security.

In conclusion, Dr. Dawit expressed hope that this training would mark the beginning of a larger initiative to leverage Africa's own capacity to strengthen its potential and tackle its challenges. He emphasized the importance of engaging higher-level policymakers and raising awareness on climate change and other critical issues. Finally, he thanked the Government of Botswana and extended his best wishes for effective deliberations and successful training.

OPENING REMARK



Dr. Charity Kruger, Executive Director of Research and Development at the National Agricultural Research and Development Institute (NARDI), on behalf of the Government of Botswana, the Ministry of Agriculture, and the leadership of NARDI, began her opening remarks by warmly welcoming everyone to Botswana. She expressed gratitude to the training organizers for focusing on the critical issues of climate change and climate information, which are crucial to agricultural and food security experts.

Dr. Kruger encouraged participants to make the most of the program and emphasized the importance of the knowledge they would gain, which would help them return to their institutions better equipped to advise leaders and develop interventions to make their countries more resilient to climate change. With these remarks, Dr. Kruger officially declared the workshop open and wished everyone an enjoyable and productive experience in Botswana.



OVERVIEW OF THE TRAINING WORKSHOP



Dr. Teferi presented the overview of the training by emphasizing that this workshop is part of a capacity-building initiative focused on climate basics, CIS, EWS, and related topics for agriculture and climate experts. He reiterated the workshop's key objective: building the capacity of agricultural experts in the region by equipping them with essential knowledge and skills in climate basics and climate information. The aim is to strengthen links between regional centers and enhance the understanding of climate change risks, drivers, and impacts.

The workshop covers five key areas: climate basics, CIS, EWS, DRM, and climate and gender, with each day focusing on one topic. Expected outcomes include increased understanding of climate change, CIS, and access to tools like map rooms and crop models. By the end, participants will know where to find useful climate tools and how to apply them. Dr. Teferi also outlined the daily schedule.

Dr. Teferi concluded by inviting participants to consider this workshop as phase one and expressed the hope that future phases would follow to continue building on the knowledge gained. He encouraged everyone to actively participate, share their expertise, and collaborate in making the training workshop successful.

CLIMATE BASICS

Dr. Tufa Dinku, Senior Research Scientist at the International Research Institute for Climate and Society, Columbia Climate School, presented on climate basics, highlighting the following key points:

- o **Overview of CRMAE curriculum:** the curriculum was developed with input from several countries, including Ethiopia, Kenya, Zambia, Ghana, Mali, and Senegal. The curriculum has four modules: climate basics, climate information products and tools available for agriculture (ENACTS, ag-data hubs, and others), climate-sensitive agricultural decisions, and integrating climate services into agricultural extension (communication).
- o **Basic concepts of weather, climate, and climate variability:** Weather describes the current atmospheric conditions, such as temperature, wind, and humidity, at a specific time and place. In contrast, climate refers to the long-term average of weather patterns, typically measured over 30 years or more, for a particular region. Climate variability encompasses short-term fluctuations or deviations from the average climate,



such as seasonal anomalies or year-to-year variations. On the other hand, climate change signifies persistent long-term changes in climate patterns, often driven by natural processes and human activities. These shifts have profound implications, including altering agricultural productivity, modifying ecosystems, and intensifying extreme weather events.

- o **Importance of spatial and temporal variation in understanding climate:** spatial variation of climate involves studying differences across different geographical locations, while temporal variation looks at changes over time through time series analysis (e.g. trends, variability, cycles, etc.). One of the fundamental global factors involved in spatial variation of climate is the solar energy imbalance, where the equator received the most direct sunlight, leading to higher temperatures, while the poles receive less, resulting in colder temperatures. Additionally, local factors such as topography, land-water distribution, land cover, and surface characteristics shape a region's climate.
- o **Climate variability:** is primarily driven by factors such as sea surface temperatures (SSTs) and global phenomena like the El Niño-Southern Oscillation (ENSO). **ENSO** involves changes in sea surface temperatures in the central and eastern Pacific Ocean. ENSO has three phases: neutral, El Niño, and La Niña, with El Niño being the warming phase and La Niña the cooling phase. These phases significantly influence weather patterns, leading to floods, droughts, and extreme temperatures worldwide. The main oceanic regions used to monitor ENSO are NINO3 and NINO3.4. The Indian Ocean Dipole (IOD) is another key factor, with the Dipole Mode Index (DMI) used to track its variations. These oceanic phenomena result in interannual variability, which affects the climate on a range of temporal and spatial scales.
- o **Climate data, information, and analysis:** it involves gathering various measurements (e.g., temperature, air pressure, humidity) from different sources such as meteorological stations, satellites, and ships. Moreover, re-analysis data, which are produced through a complex process that combines past observations with Numerical Weather Prediction (NWP) models, is a valuable resource for climate research, monitoring and applications. While raw data is valuable, it is often transformed into actionable climate information through analysis. This allows for more specific insights, such as identifying trends in extreme weather events. Different types of climate data, including historical records, real-time monitoring, and forecasts, serve specific purposes, from planning and disaster management to operational risk management.
- o **Climate forecasting** includes weather forecasts (short-term) and seasonal forecasts (long-term), which depend on different conditions. Weather forecasts focus on initial conditions, while seasonal forecasts rely on external boundary conditions, particularly SSTs. Seasonal forecasting aims to predict average climate conditions over a season, such as temperature and precipitation. It utilizes both statistical and dynamical methods. The statistical method relies on historical climate data to identify recurring patterns and relationships between different variables, whereas the dynamical method uses numerical climate models that simulate the Earth's atmosphere, ocean, land surface and ice. In recent years, there has been a trend towards combining statistical and dynamical methods to produce seasonal forecasts.



- o **Barriers to climate forecasting:** Several challenges hinder the effective utilization of climate forecasts. One of the main obstacles is scientific limitations, where forecasting models may not be advanced enough to ensure accurate predictions. Another issue is the use of inappropriate formats for presenting forecasts, which may not align with the users' needs. Additionally, external constraints, such as institutional barriers or logistical challenges, can limit the ability of users to act on the forecasts. Finally, communication challenges arise when different forecasts provide conflicting predictions, leading to confusion about which forecast to trust.

CLIMATE INFORMATION SERVICES

Dr. Tufa's presentation on Climate Information Services centered on two key topics: an Overview of Climate Services and The ENACTS Approach.

- o **Overview of Climate Services:** the Global Framework for Climate Services (GFCS) defines a climate service as "a decision aid derived from climate information that assists individuals and organizations in society to make improved decision-making." Similarly, the Climate Services Partnership (CSP) defines it as the production, translation, transfer, and use of climate knowledge and information in climate-informed decision-making and climate-smart policy and planning.
- o **Pillars of climate services:** include the generation, translation, transfer/communications, and application/use of climate information. These services operate at global, regional, national, and local levels.
- o **Global Framework for Climate Services (GFCS):** is crucial in accelerating and coordinating the technical and scientific efforts needed to improve climate-related outcomes across the pillars of aforementioned climate services.
- o **National Framework for Water, Weather, and Climate Services (NFWWCS):** was introduced as an institutional mechanism to improve collaboration among national institutions to co-produce, tailor, deliver, and use science-based climate predictions and services. Several National Meteorological and Hydrological Services (NMHS) have begun implementing this framework.

ENACTS APPROACH

The second part of Dr. Tufa's presentation focused on the ENACTS (Enhancing National Climate Services) Approach.

- o **Overview of ENACTS:** it aims to improve climate information availability, access, and use. IRI collaborates with NMHS to quality-control available station data and integrate them with satellite and reanalysis products. The approach's primary goal is to create reliable climate information for informed local decision-making.



- o **Pillars of the ENACTS:** enhancing access, promoting use, and improving data availability. To enhance access, ENACTS performs data analysis and visualization. It promotes use by engaging end-users to raise awareness, build capacity, and involve them in product development. Improving data availability focuses on building NMHS capacity, ensuring the quality control of station data, and combining these with proxy data to enhance seasonal forecasting.

Several ENACTS tools were introduced during the presentation:

- o **The Climate Data Tool (CDT):** this tool has six main modules—data preparation, data quality control, generation of ENACTS data, evaluation, analysis, and visualization.
- o **The Automatic Weather Stations Data Tool (ADT):** designed to manage data from various Automatic Weather Station (AWS) systems, ADT organizes, quality-controls, processes, and visualizes AWS data, enabling real-time monitoring.
- o **The Climate Predictability Tool (CPT), Python version (PyCPT):** this tool optimally combines multiple models, can be calibrated with local data, to generate full probability forecasts.
- o **The IRI Data Library:** this is a powerful platform for storing, analyzing, and visualizing climate data, installed at NMHS for secure data sharing. It is the critical component of ENACTS for making climate information products available to the public while safeguarding the underlying data. It is a platform for secure data sharing within and outside the NMHS.
- o **ENACTS Maprooms:** an interactive online climate analysis and visualization portal.
- o **Capacity development:** This component focuses on training NMHS staff and end-users to use the provided tools. It also includes developing a curriculum to support this training.

The key outputs of ENACTS include creating a 40-year climate time series for every 4km grid across each country and providing data in areas without weather stations. The initiative also introduced powerful tools like CDT, ADT, and PyCPT for NMHSs and other users. It enabled unprecedented online access to climate information products, addressing many users' needs and partially overcoming data access challenges. Additionally, ENACTS built capacity at NMHS and within user communities, particularly in the agriculture sector, through the CRMAE curricula.

DEMONSTRATION OF ENACTS

Dr. Tufa Dinku began the demonstration by focusing on Madagascar. He highlighted the map room, which displayed various categories like climate, climate and water, climate and health. He explained the three available tabs under the climate section: climate analysis, climate monitoring, and climate forecast. Climate analysis allowed users to view daily, ten-day, monthly, and seasonal trends for rainfall or temperature over the past 30-40 years.



These trends could reveal increasing or decreasing temperatures, drought probabilities, or flood risks. The climate monitoring tab, which was supposed to be updated every 10 days, showed the latest data, but Madagascar's updates had stopped in January, while Ethiopia's data was current.

In Madagascar, Dr. Tufa demonstrated how users could analyze monthly rainfall by selecting specific months or regions. This feature allowed users to see rainfall patterns and climatology across different areas. He also showed how to zoom in on specific locations to extract temporal data, which displayed trends for rainfall or temperature over time. He also showed how to run an extreme rainfall analysis, calculating the probability of exceeding specific rainfall amounts during certain seasons. He also explained how users could reverse the analysis to see the probability of rainfall not exceeding a given threshold, which would be useful for agricultural planning. Throughout the demonstration, he emphasized how the tool allowed users to evaluate rainfall and temperature trends to make informed decisions, such as selecting suitable crop varieties based on climate suitability. He also demonstrated the tool's daily analysis feature, which could assess agricultural parameters like dry spells or rainfall intensity, aiding in crop planning.

Discussion points

- o In Kenya, providing localized climate information to farmers but integrating data has been challenging, especially with access to official data. After years of struggles, they have started adjusting daily data and offering farmers advice every season. They aim to improve by offering weekly or monthly forecasts and they are evolving the extension system digitally, appealing to youth and improving access to information.
- o Tanzania uses seasonal forecasts and collaborates with stakeholders to translate these into actionable advice for agriculture and other sectors. The Meteorological Office shares forecasts through media, websites, and social media, ensuring that agricultural officers get regular updates. They also work closely with ICPAC for additional seasonal forecast insights.
- o Participants noted that while they receive seasonal forecasts, like for the OND period, and collaborate to interpret them for farmers, the one-month lead time is insufficient. They also rely on regional monthly forecasts, but still find the time inadequate to fully prepare farmers.
- o Most farmers have cell phones, so timely updates are essential. For instance, when forecasting above-average rainfall, it's crucial to specify timing as conditions can change rapidly. Recently, conflicting forecasts—one predicting neutral weather and another La Niña—caused confusion, highlighting the need for frequent updates.
- o Farmers often use initial forecasts to make investments, so if we later update the forecast or shift direction, it creates challenges. The gap between early forecasts and updated advice can leave farmers in difficult positions if not handled properly.
- o ICPAC produce monthly forecasts that cover both the current month and the upcoming months, including the seasonal outlook and forecasts are updated forecasts every month. Additionally, ICPAC issues weekly forecasts that provide expected rainfall and



temperature conditions for the week ahead. For certain rainfall seasons, ICPAC release forecasts 2 to 3 months in advance. For example, we issued an alert in July for the October to December rainy season.

GENDER AND CLIMATE INFORMATION SERVICES

Joyce Jelagat, a gender and climate expert from ICPAC, presented on gender and climate information services. The core points discussed were

- o **Concept of gender:** it refers to the socially constructed characteristics of men, women, boys, and girls, which differ from one community to another. It is important to note that gender is not limited to biological sex but also includes the social and cultural roles, behaviors, and responsibilities assigned by society to different gender groups.
- o **Gender sensitivity:** is the ability to recognize and address the differences and inequalities between gender groups. It involves acknowledging these differences when designing and implementing strategies and actions, ensuring that each group's unique needs and experiences are considered. By being gender-sensitive, we can create policies and services that are more inclusive and equitable.
- o **Social inclusion:** is the process of ensuring that equal opportunities are available for everyone, regardless of gender, socioeconomic status, or background. The aim is to ensure everyone has a voice and an opportunity to participate in decision-making processes.
- o **Intersectionality:** looks at how different characteristics such as gender, race, class, sexual orientation, and physical ability intersect to influence a person's vulnerability or privilege. It highlights the complexity of identity and how various factors can contribute to marginalization or empowerment.
- o **Gender mainstreaming:** this approach ensures policymaking and program design consider the interests of all gender groups, including men, women, youth, the elderly, and people with disabilities. Different societies have different vulnerabilities, and gender mainstreaming seeks to address these by incorporating gender perspectives into the design, implementation, and evaluation of policies and programs.

In her explanation of the connection between climate services and gender, she presented two scenarios:

Scenario 1: Samuel is a 60-year-old Maasai man living in a semi-arid region with his wife and ten children, with an 11th child on the way. As a pastoralist with no formal education, Samuel faces challenges in accessing services such as schools and hospitals, which are located far from his home. It is predicted that there will be above-normal rainfall in this region, yet Samuel's family has not received any climate or weather information.

Some of the reflections by participants are summarized as follows



- o One participant pointed out that Samuel, having lived through various seasons, might rely on indigenous knowledge to predict weather patterns based on natural signs, such as cloud movements or winds. While this traditional knowledge can be helpful, it may not be sufficient without formal climate services.
- o Another participant highlighted that Samuel's family might struggle to access formal climate services due to their distance from urban centers, lack of education, and limited technological access.

Scenario 2: a poor, urban community living in an informal settlement. The residents, who rely on casual labor to survive, have limited access to well-equipped schools and health centers, and the community also struggles with issues such as GBV. In this context, the area is predicted to experience above-normal rainfall, but the community has not yet received this information.

Some of the reflections by participants is summarized as follows

- o One participant observed that this community faces high exposure to climate risks and low capacity for adaptation due to poverty and poor infrastructure. Even if they receive climate information, their response ability may be limited.
- o Another participant highlighted the potential increase in GBV during extreme weather events, as families might be confined indoors for extended periods. The lack of community centers and safe spaces exacerbates the situation.
- o Others noted that disruptions in the normal functioning of families, especially in informal settlements, can prevent men and women from accessing necessary climate information and services.

Several important factors affect how different groups, especially marginalized communities, can access and use climate information. The challenges include;

- o **Literacy barriers:** low levels of education hinder understanding of complex climate terms, such as "below normal rainfall," especially in rural areas where these terms may not be meaningful in practical terms.
- o **Lack of awareness:** marginalized communities often lack knowledge of early warning services, making it impossible for them to seek or access information they don't know exists.
- o **Cultural and traditional norms:** in certain communities, such as pastoralist societies, women are restricted from attending public forums due to cultural norms, preventing them from receiving climate information shared at community meetings.
- o **Technological barriers:** many communities have limited access to mobile phones, radios, or the internet, which are common channels for disseminating climate information, reducing their ability to receive updates.



- o **Time constraints:** gender roles and responsibilities affect how and when men and women can access climate information. For example, women may miss critical weather updates while engaged in tasks like gathering firewood, and those with casual jobs may not have time to access broadcasts or alerts.
- o **Limited access to resources:** climate information is only useful if people can act on it. Poverty, lack of financial resources, and limited access to land make it difficult for people to take actions like relocating or adjusting agricultural practices in response to climate warnings.
- o **Land access challenges:** in many African societies, women have restricted land ownership rights, limiting their ability to make timely agricultural decisions based on climate forecasts.
- o **Financial constraints:** without sufficient collateral, such as land, many people in marginalized communities are unable to secure loans needed to improve crop production or safeguard their livelihoods against climate risks.

Discussion points

- o Prioritize social inclusion in our climate services, especially considering the challenges in Eastern and Southern Africa. It's important to ensure fair distribution and accessibility to all people.
- o Develop a policy framework that applies across multiple sectors, not just agriculture. This framework should include action plans that guide policymakers, practitioners, and communities on gender and social inclusion.
- o Policies and strategies must move beyond theoretical frameworks and literature. Ground-level actions and diverse voices are crucial to making real change.
- o Adopt strategies for improvement: (i) comparative research: researchers often engage communities for data collection but don't always leave something meaningful behind. We need to ensure communities are custodians of their own data and benefit from it and make sure to train local people to understand climate products (ii) Mobile Services: drawing inspiration from mobile health clinics, we could develop mobile meteorological or climate services that regularly provide information to remote communities, similar to how mobile health services operate which would bridge the gap in climate services accessibility.

IMPLEMENTATION OF CLIMATE SERVICES IN EASTERN AFRICA

Dr. Hussen Seid, Climate Modeling Expert at ICPAC, presented on the implementation of climate services in Eastern Africa, covering the following key points:



- o **Climate monitoring and data collection:** ICPAC's climate monitoring and forecasting rely on data from national meteorological services combined with satellite information. However, the limited number of stations providing data in some countries remains a challenge.
- o **Climate products:** ICPAC produces various products, including 10-day summaries, seasonal forecasts, and rainfall analysis. Weekly forecasts are issued every Monday, while longer-term seasonal outlooks are developed using statistical and dynamical methods. ICPAC utilizes global forecast models from centers like ECMWF which are downscaled and calibrated for the region.
- o **Dynamical downscaling:** high-resolution weather models are run over Eastern Africa, producing forecasts at a 10-kilometer resolution. This detailed forecasting includes products such as total rainfall, rainfall anomalies, and probability-based predictions, providing valuable insights for users.
- o **Temperature products:** for temperature, the focus is on averages, while for rainfall, totals are important. ICPAC also monitor temperature anomalies, such as positive or negative deviations. Heat stress is an emerging hazard in the region. So, ICPAC provides temperature forecasts that highlight areas expected to face extreme conditions in Eastern Africa.
- o **Statistical downscaling methods:** three methods for statistical downscaling to produce forecasts are used. The final seasonal forecast is generated using multiple prediction systems. Many users rely on probabilistic forecasts, but low probabilities often pose challenges for decision-making. At ICPAC shifted to an objective forecasting methodology in 2019. This method, recommended by the WMO, produces reproducible, verifiable forecasts that are traceable. This approach allows for more relevant products, such as exceedance forecasts, which improve forecast reliability.
- o **Monthly rainfall distribution:** in addition to seasonal forecasts, ICPAC also provide forecasts for each month. Even if a seasonal forecast predicts above-normal rainfall, there can be significant variation in rainfall from month to month.
- o **User-specific forecasts:** users have indicated that traditional tercile forecasts are insufficient. To address this, ICPAC has begun producing more targeted products based on user-defined thresholds, such as the probability of rainfall exceeding 200mm, 300mm, or 400mm. This is crucial for agriculture, where specific amounts of rainfall are needed for crop growth.
- o **Start of rainfall season:** another key product for decision-making is the forecast for the onset and cessation of the rainfall season. Users need to know the expected dates and the probabilities of both onset and cessation to plan agricultural and other activities effectively. The onset of the rainy season can be defined using daily forecast data from models. ICPAC consider the first day of the wet season to be when the rainfall totals at least 20mm over three consecutive days, and there is no dry spell longer than seven days in the following 21 days. Once this threshold is met, we identify it as the start of the rainy season.



- o **Dry and wet spells forecast:** ICPAC also produce forecasts for expected dry and wet spells during the rainy season. This includes predicting the maximum duration of these spells and when they will occur. Additionally, we're working on applying machine learning to improve our forecasts, particularly for mid-season predictions where current models lack accuracy.
- o **Sub-Seasonal forecasting:** there's a gap between weekly and seasonal forecasts, referred to as sub-seasonal forecasting, which is crucial for agriculture. This detailed outlook is valuable for sectors like agriculture that need more specific time frames within the season.
- o **Forecast verification:** after the season ends, ICPAC verifies forecasts to assess their accuracy. This is an advantage of the objective forecasting approach, as it produces digital forecasts that allow for statistical verification which enables to evaluate which models and methods performed best and improve our future forecasts. In contrast, consensus-based forecasting can only be visually verified by comparing predictions to observations, lacking the statistical depth of digital forecast analysis.

CO-PRODUCTION PROCESSES DURING THE GREATER HORN OF AFRICA CLIMATE OUTLOOK FORUM (GHACOF)

Dr Masilin Gudoshava, research scientist at ICPAC, talked about co-production processes during the Greater Horn of Africa Climate Outlook Forum (GHACOF). She touched on the following key points during her talk

- o **Co-production:** it involves bringing together individuals from different fields of expertise with the goal of creating a product that is easily usable. Co-production starts with identifying key stakeholders and building partnerships. Sometimes users may request something that producers cannot deliver, so early communication is essential to stay aligned. Both producers and users work together to identify gaps and co-develop solutions. Developing evaluation metrics that meet both scientific and user needs is crucial.
- o **Methods of co-production:** there are different types of co-production methods, ranging from consultative to immersive. More immersive methods often require more resources, so it's important to consider resource availability from the start.
- o **Climate Outlook Forums:** there are 21 regional climate programs worldwide, and the idea of COF to avoid conflicting forecasts from different centers, leading to confusion. To date 68 sessions have been held with the objective of reviewing the previous season's performance, providing forecasts for the upcoming season, formulating mitigation measures, and sharing experience.
- o **GHACOF process:** at ICPAC, COFs are conducted three times a year, corresponding to the three major rainy seasons. Capacity-building workshops for member states are held. The main COF event gathers stakeholders from various sectors to engage in co-production and provide feedback.



- o **Dissemination:** climate information is disseminated through various platforms, including media, and in multiple languages like French, Swahili, Amharic, and Arabic, to reach a wider audience. The information cascades from regional to national levels, where NCOFs engage local stakeholders, and at the community level, indigenous knowledge is integrated.
- o **User interface platforms:** platforms such as the Food Security and Nutrition Working Group for Eastern Africa bring together organizations from UN agencies, research institutions, NGOs, and donors. In this group, the focus is on climate, livestock, and food security, providing advisories based on upcoming climate conditions. If an extreme event, such as flooding, is predicted, alerts are issued to guide humanitarian responses. In addition to this, ICPAC is also focusing on establishing similar frameworks for the water and energy sectors.
- o **Training journalists for better climate communication:** to further strengthen the communication of climate information, journalists are being trained on how to use and communicate climate data. There is an online course available that provides this training, and it's accessible in multiple languages including English, French, Kiswahili, and Arabic. This initiative is helping to ensure that the public receives accurate and understandable climate information through the media.

SOCIO-ECONOMIC BENEFITS OF CIS IN AGRICULTURE AND FOOD SECURITY

Dr. Masilin Gudoshava presented on the socio-economic benefits of CIS in agriculture and food security, highlighting several key points throughout her talk.

- o **Overview of food security in SSA:** over the past five years, there has been a clear and steady rise in food insecurity across several regions, with Eastern, West and Central Africa being the most affected where most communities rely on rain-fed agriculture, which is highly vulnerable to climate variability.
- o **Decision-making in climate services:** effective decision-making is a crucial benefit of using climate information. Across the spectrum of available climate data—from nowcasting to long-term climate change projections—there are numerous ways these services inform critical decisions. Along the agricultural value chain, different climate information products are needed at various stages. Seasonal forecasts, sub-seasonal forecasts, and early warnings for extreme weather events all play a role. For example, one tool used in Eastern Africa is the Agricultural Watch, which provides data on productivity and early warnings for farmers.
- o **Economic benefits of climate information:** climate services have a significant economic impact. For example, looking at Kenya, the projected losses in GDP without the use of climate services are considerable, but by integrating climate information, GDP losses can be mitigated. Similarly, in Uganda, it is estimated that up to \$3 billion in GDP losses could be avoided with the use of climate services.



PILLARS OF EARLY WARNING SYSTEM (EWS)

Dr. Ernest Afiesimama, Regional Programme Manager at the World Meteorological Organization, presented the rationale for investing in EWS and strengthening disaster preparedness in Africa. His talk focused on the following core points:

- o **Definition and role of EWS:** EWS involve tools and processes designed to detect and communicate potential hazards before they escalate into disasters. Their purpose is to provide timely and actionable information to help communities prepare for and mitigate the effects of emergencies. These systems aim to reduce loss and damage from hazards such as floods, droughts, and heatwaves by focusing on hazard detection, data collection, and communication of warnings.
- o **Context for EWS in Africa:** Africa is experiencing warming faster than the global average despite its low contribution to global emissions (3-4%). The continent faces increasingly erratic precipitation, with above-average rainfall in the Gulf of Guinea contrasting with severe deficits in North, East, and Southern Africa. Rising sea levels, particularly around the Red Sea, threaten coastal communities and widespread flooding in East. Additionally, severe droughts and heatwaves, especially in Northern Africa, are worsening food insecurity, public health, and infrastructure vulnerabilities.
- o **Projections and investment in EWS:** looking ahead, if no mitigation measures are taken, up to 118 million people in Africa could be exposed to extreme droughts, floods, and heatwaves by 2030. The growing risk of extreme climate events underscores the urgency for African countries to adopt and invest in comprehensive EWS to safeguard communities and economies. Key areas of focus of investment include modernizing observation networks, improving forecasting and modeling capabilities, and establishing robust communication channels to disseminate early warnings. Additionally, enhancing training, building capacity, and fostering collaboration will be vital.
- o **Data and science translation into policy:** translating science into actionable policy is essential. Authorities are striving to change their approach, and while data is available, a caveat remains regarding the need to enhance data collection and analysis to inform policymakers more effectively.
- o **Key components of EWS:** the components are hazard detection, communication, response planning and feedback mechanisms.
- o **Socio-economic importance of early warning systems:** in 2022, the UN Secretary-General emphasized the importance of early warning and early action systems, which are cost-effective ways to mitigate disaster impacts. Economic analysis shows that for every \$1 invested in early warning systems, the return can be tenfold. However, 60% of Africans are not covered by these systems, which is a significant risk.
- o **Pillar 1 - Disaster Risk Knowledge:** serves as the foundation of an effective Early Warning System (EWS) by gathering and analyzing data on past disasters and vulnerabilities specific to different communities. This pillar promotes inclusive approaches by integrating local and indigenous knowledge and using clear, accessible



language to communicate risks. It also highlights the importance of maintaining disaster loss databases, utilizing sector-specific data, and scaling innovations to strengthen disaster risk understanding and support informed decision-making.

- o **Pillar 2 - Detection, Observation, Monitoring, Analysis, and Forecasting of Hazards:** emphasizes continuous observation, monitoring, and forecasting of hazards. This pillar relies on accurate data from various sources, including ground-based stations and satellites, to eliminate gaps that could compromise forecast accuracy. Forecasts should present quantitative data and highlight potential impacts—like property damage from severe weather—to enable community preparedness. Expanding and maintaining observational networks requires collaboration and funding to ensure infrastructure meets global standards, supported by initiatives like the Systematic Observation Financial Facility.
- o **Pillar 3 – Warning dissemination and communication:** is focused on ensuring early warning messages reach at-risk communities quickly and effectively. This pillar employs multiple communication channels, such as mobile phones, radio, TV, social media, and satellite networks, to maximize coverage. The Common Alerting Protocol standardizes emergency alerts across platforms, enhancing consistency and clarity. Collaboration with mobile operators and national telecom networks is essential to ensure timely message delivery, with the goal of enabling communities to receive alerts promptly and take necessary actions.
- o **Pillar 4 – Preparedness to response:** emphasizes strengthening preparedness and response capacities to reduce disaster impacts. This involves creating detailed community response plans, such as evacuation routes, shelters, and emergency services that cater to local needs. Regular drills and simulations are essential to test community readiness and ensure people know how to respond when disasters occur. Effective preparedness requires strong coordination among government agencies, NGOs, and local communities, building disaster response capacity at all levels—from local to national—to ensure communities can manage and mitigate disaster effects effectively.
- o **Coordination and implementation:** efforts are underway to develop coordination mechanisms at both national and continental levels to strengthen EWS. This includes stakeholder mapping, gap analysis, and creating roadmaps for implementation, all of which will help ensure the sustainability of these systems at local, national, and regional levels.

CROP CAPABILITY TOOL FOR AGRO-ADVISORY FOR CSA IN SUB-SAHARAN AFRICA

Dr. Bradwell Garanganga from Digitron presented the Crop Capability Tool for Agro-Advisory, focusing on Climate-Smart Agriculture (CSA) in Sub-Saharan Africa. He emphasized the following key points:



- o **Global economic cost of climate disasters:** CIS have the potential to generate returns as high as ten times the initial investment, with agriculture being a major beneficiary of a robust CIS. In developing nations, climate-related events typically result in losses amounting to 10-14% of GDP. The economic impact and human toll of extreme weather remain severe, as illustrated by recurring devastations from tropical cyclones and droughts across the SADC region and other parts of Africa.
- o **Agricultural productivity in SSA:** in SSA, agriculture is central to both food security and employment, contributing approximately 15% to the region's GDP, however, productivity remains low.
- o **Impacts of climate variability and changes in SADC:** climate change and variability continue to impact food security in SADC, with around 60 million people (17% of the population) facing food insecurity in the 2023/24 season. Losses in crops and livestock could have been reduced through timely use of CIS based on SARCOF forecasts.
- o **CIS-Based DST:** climate variability and change impact 70% of agricultural production costs, making DST in Agriculture and Food Security essential for guiding communities toward optimal productivity. Developing DSTs can contribute to greater efficiencies in agricultural production systems, improve food security by enabling cost-effective import and export decisions, reduce the impacts of hydrometeorological hazards, and enhance the contributions of climate-sensitive sectors to sustainable economic growth.
- o **Development of DST:** the tool integrates SCF and crop simulation models to provide long-term crop yield forecasting capabilities. Key models include the Climate-Agriculture Modelling and Decision Tool (CAMDT) and the Decision Support System for Agrotechnology Transfer (DSSAT) for crop yield prediction. CAMDT, an open-source tool, drives DSSAT by combining data on crops, soils, and weather with crop models and applications to simulate various crop management strategies. Digitron has also developed an app that simplifies data management, especially for non-rice cultivars.
- o **Actionable information & products:** decision criteria will be established to help users take proactive actions based on seasonal climate forecast probabilities, enabling them to mitigate losses or enhance productivity. The model will guide users in choosing strategic actions triggered by SCF probabilities up to five months in advance, with some yield forecasts available early enough for input procurement and planning. During the season, users can apply CIS for tactical adjustments.
- o **Probability of exceedance curves:** used as input for DSS, that guide farmers in deciding the extent of land to allocate to specific crops each season based on available SCF and crop yield predictions. These yield exceedance curves adjust proportionally to each forecast tercile category, providing tailored insights for each crop cultivar. The model can also compare yields for the same cultivars across different planting dates or management practices, enhancing decision-making for optimal farm management.
- o **Scenario generation:** the tool can be adapted to run for a variety of crop cultivars and seed varieties if available, enabling the generation of diverse scenarios. These scenarios include insights on different crop cultivars, planting dates, years, and management systems, allowing users to analyze seasonal characteristics across these



variables. This versatility highlights the Crop Capability Model's significant value as a decision-support tool.

- o **Next steps:** successful DSS implementation for agriculture and food security requires policies and investments. Continuous model improvements should extend lead times, expand spatial scales, and support a web-based app for accessibility. Training of trainers is vital, along with extending studies to climate scenarios and scaling across more African countries to enhance agricultural resilience.

EXPERIENCE SHARING

Climate Education

Professor Berhanu Belay shared insights on integrating climate knowledge products (CKP) into university curricula in Ethiopia, addressing the following key points:

- o **Role of education in climate mitigation and adaptation:** education plays a critical role in climate change mitigation and adaptation. However, it is frequently overlooked by policymakers. Data from Africa's 53 countries show that, compared to other sectors like agriculture and food security, education receives minimal focus in climate adaptation strategies. This highlights the urgent need to prioritize education in our efforts to combat climate change.
- o **Curriculum development and integration of CKP:** the curriculum is foundational in education, shaping what students learn and how they learn it. It includes the content delivered, the teaching methods used, and assessments that measure student competency. In Ethiopia, a gap analysis of agricultural programs revealed significant deficiencies in climate-related content, especially in animal science. Addressing these gaps involves collaboration with universities and the Ministry of Education, forming committees to review and integrate CKP into academic programs.
- o **Approaches to integrate CKM:** several strategies were employed to integrate climate knowledge into Ethiopian universities including initiating new programs, curriculum review and quick revamps, student engagement (through climate-related student clubs) and customized training.
- o **Integrating climate education across levels:** AICCRA is working with the Ministry of Education to embed climate education across all educational levels, from kindergarten to higher education.
- o **The role of media in climate education:** media plays a crucial role in raising public awareness about climate issues. Empowering local media to communicate climate-related information effectively is very important. Training journalists on climate topics ensures that they can accurately and persuasively convey these messages.
- o **Challenges in curriculum integration:** despite progress, integrating climate knowledge into curricula faces several challenges. Curricula often remain static when



they should be dynamic, adapting to evolving needs and evidence. Additionally, there is resistance to interdisciplinary integration, such as including climate content in specialized fields like animal science.

Role of ASARECA in Enhancing Agricultural Resilience to Climate Change in ECA

Moses Odeke presented the role of Association for Strengthening Agricultural Research in Eastern and Central Africa (ASARECA) in enhancing agricultural resilience to climate change in Eastern and central Africa. He touched on the following major points.

- o **Overview of ASARECA:** it is a non-profit intergovernmental organization established in 1994 by 10 founding NARIs. Its membership has grown to include 15 countries. ASARECA's mandate is to coordinate and mobilize regional collective action, financial and human resources to implement priority AR4D interventions across the ECA subregion.
- o **Partners:** at the national level, ASARECA collaborates with a wide range of stakeholders, including national agriculture research and extension systems, agricultural institutes, and various groups such as extension agents, farmers, youth, policymakers, educators, private sector entities, civil society organizations, and women. At the regional level, ASARECA partners with regional economic communities, IGAD, ESC, etc., to broaden its impact.
- o **ASARECA thematic areas:** include capacity strengthening and integration, agricultural transformation technologies and innovations, creating an enabling policy environment, developing functional markets and transformative institutions, and managing knowledge and information.
- o **Major investment areas in member countries:** the areas include promoting regionally prioritized TIMPs along selected value chains, capacity building and fosters partnerships at individual, organizational, and systemic levels while coordinating policy formulation, analysis, advocacy, and reforms at national and regional levels. It also aims to improve access to national and regional markets for key commodities, address the impacts of climate change through adaptation mechanisms, enhance intra- and inter-country trade, and facilitate CKM.
- o **ASARECA approaches to enhance resilience:** support generation of CSA/CSL TIMPs; scale up CSA/CSL technologies and tools and increase access to CSA/CSL technologies, knowledge and information on CSA/CSL technologies and insurance schemes.
- o **Capacity strengthening:** ASARECA strengthened capacities of stakeholders in various thematic areas through targeted trainings. Through these 1,387 stakeholders trained and applying skills.
- o **Mapping climate-relevant data sources:** a work to identify relevant data sources and determine how to access their datasets is done and successfully mapped approximately 25 different sources of satellite data, of which 20 were available through open data sources, while two required payments for access.



- o **Interventions and findings:** climate change interventions have been identified across different countries, Kenya leads with 63 initiatives, followed by Uganda and Tanzania, while countries like Eritrea have only two. Of the CSA interventions, the majority focus on food security, with about 32% addressing adaptation and only 10% on mitigation strategies.
- o **Virtual irrigation academy:** it is designed to mitigate the effects of climate change on water shortages. The tools developed can help farmers reduce irrigation water usage by up to 50%, thereby improving water availability for livestock. The moisture indicators guide farmers on when to irrigate, significantly reducing their water consumption.
- o **Potential areas for collaboration:** include developing strategies to support the dissemination and scaling of professional services and focusing on capacity strengthening to help stakeholders utilize climate information services effectively. ASARECA aims to engage high-level policymakers for their buy-in, which is crucial for the success of scaling up initiatives.

Role of CCARDESA in Enhancing Agricultural Resilience to Climate Change in SADC countries

Barthlomew Chataika presented the role of CCARDESA in enhancing agricultural resilience to climate change in Southern Africa. He touched on the following major points.

- o **Overview of CCARDESA:** it covers the 16 member states in the Southern part of Africa, which has approximately 370 million people. It is established in 2010 and officially launched in 2011, is a subsidiary of the SADC. Its primary purpose is to coordinate agricultural research and development across the region.
- o **Mandate:** focus on six key thematic areas: enhancing agricultural productivity; ensuring food and nutrition security; building resilience to emerging agricultural risks; promoting commercialization; advancing inclusion of women, youth, and vulnerable groups; supporting knowledge management, communication, and policy; and strengthening capacity development.
- o **Priority areas:** CCARDESA priority areas for research development include availability of quality and improved seeds (breeding), pest and disease, markets, impacts of climate variation on crop production, soil fertility, and capacity building.
- o **COVID-19 response initiatives:** during COVID-19, CCARDESA responded rapidly by launching projects to address food security in four countries. This included setting up climate-smart facilities and enhancing resilience through region-specific interventions.
- o **AICCRA project:** through this project a CSA model in Chipata, Zambia has been established to implement a solar-powered irrigation system, introduce sustainable intensification technologies to support CSA-aligned agricultural production, and developing woodlots aimed at reducing deforestation. The project also helps in capacity building efforts.



- o **CAADP-XP4:** focuses on strengthening capacity and enhancing the automation of MEL systems to improve efficiency and effectiveness. It promotes innovative partnerships and public-private partnerships to drive progress in the agricultural sector. A key priority is the development of a regional action plan aimed at integrating climate science and technology indicators within National Agricultural Investment Plans. Additionally, the program addresses priority areas in agricultural research and has developed Knowledge Management strategies.
- o **FSRP:** the program emphasizes Knowledge Management through innovative technology dissemination, utilizing digital tools, animations, and FFS. It also supports the development of RCoLs to strengthen regional expertise and capabilities. Additionally, the program prioritizes policy harmonization and soil health.
- o **Knowledge products:** more than 80 knowledge products developed under different programmes- GIZ/ICKM, APPSA, ACCRA.

National Agricultural Research Organization Uganda

Angela Asachi highlighted recent efforts by Uganda's National Agricultural Research Organization (NARO) to advance climate-resilient agricultural practices. She discussed the following key points:

- o **Source of real time information:** through the MoU signed between NARO and the National Meteorological Authority, it has been agreed to jointly utilize one of the weather stations, which transmit data in real-time to both organizations, allowing us to respond quickly with data-driven solutions.
- o **Utilization of the data:** the data helps to provides farmers with accurate information on planting schedules by analyzing onset and cessation dates of rainfall across Uganda's two main seasons, March–May and October–December. Field experiments were conducted using advanced statistical methods to predict optimal planting windows through identification of the onset and cessation of rainfall for various agroecological zones. This analysis allows us to create detailed maps showing rainfall patterns and seasonal timelines across different regions, tailored to semi-arid and more fertile zones alike.
- o **Output dissemination:** while valuable, the outputs were initially too technical for many farmers, so we developed an online application, the Seasonal Characteristics Tool. Accessible via desktop and Android, it offers user-friendly information and advisories tailored to specific locations and weather trends. The tool includes seasonal forecasts and crop yield data, empowering farmers to make well-informed decisions about planting and irrigation. To further support decision-making, advisories offer guidance on crop and variety selection, nutrient application, irrigation scheduling, and drought-resilient practices. Specific recommendations on crop water requirements and ideal planting conditions for different crops and seasons, aligning with Uganda's national irrigation initiatives, are also provided.
- o **Integration of AI:** AI is also integrated to analyse land management practices influencing maize yields in different agroecological zones.



- o **Crop suitability under climate-change scenarios:** this analysis assesses which crops will thrive or struggle with potential temperature increases. While complex, the aim is to integrate these insights into the tool so farmers can quickly identify viable crops for their regions.
- o **Early warning systems for pest and disease monitoring:** applying AI for early warning systems targeting pest and disease management, leveraging rainfall and temperature data to identify hotspots for issues like maize fall armyworm infestations is also being done.

Experience from Southern Africa

Dr. Tharaga Phumudzo Charles, an agrometeorologist, has had a diverse career across sectors. Starting in the private sector with Absa, he used agrometeorology to manage financial risk, offering farmers crop insurance and guidance. He then joined the South African Weather Service, supplying climate data to public and commercial clients. His interest in agricultural microclimates led him to CSIR, where he provided farmers with actionable climate insights.

Currently, Dr. Charles leads projects on crop water use and hydrology, securing \$25 million in grants, including a major apricot water-use study and Green Climate Fund support for South African agrometeorology. His international work includes agroforestry in Zambia, irrigation in Southern Africa, and wetland management with U.S. partners. At ARC, he now oversees South Africa's agricultural weather station network, ensuring critical climate data supports national agriculture.

Experience From Botswana University of Agriculture and Natural Resources

A representative from Botswana University of Agricultural and Natural Resources shared how her family's reliance on agriculture, and their limited access to climate forecasts, inspired her commitment to climate education. Observing the negative impact of climate and land-use changes on productivity in 2013, she became dedicated to helping local farmers access climate information. Early involvement with the youth-led NGO Green Society allowed her to promote environmental sustainability through eco-friendly projects supported by UNDP.

Now, she researches climate resilience in agriculture, focusing on crop impacts and adaptation strategies. She trains students and farmers on climate data, collaborates with organizations for awareness initiatives, and participates in international efforts like COP28 workshops. Her work in Botswana and Sub-Saharan Africa includes validating environmental education strategies and advocating for sustainable development in partnership with groups like UNESCO and UK Research and Innovation.

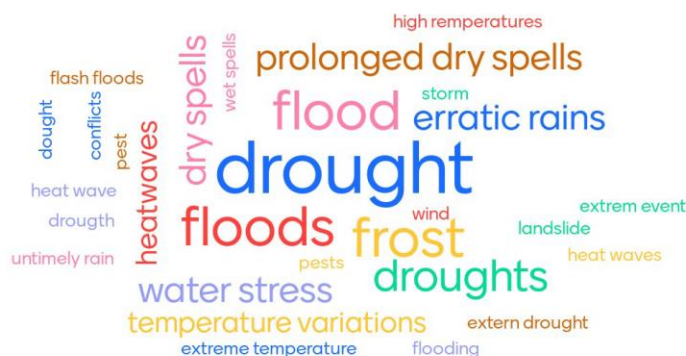
INTRODUCTION TO DISASTER RISK MANAGEMENT

Dr Ahmed Amdihun, Regional Programme Manager for DRM at ICPAC, presented an introduction to disaster risk management highlighting the following key points:



- o **The urgency of climate change:** the world cannot continue with business as usual. The reality of global warming—with rising temperatures on land and oceans—demands immediate and focused attention. This is especially critical for Africa, where many countries are experiencing alarming increases in extreme weather events, from more frequent and severe floods to prolonged, intense droughts. These patterns underscore the pressing need for decisive, coordinated action to address the escalating climate crisis.
- o **Africa's level of exposure and vulnerability:** by looking at observational data climate disasters impact high number of people in various countries and regions. Over the past two decades, nearly 50,000 lives have been lost due to these disasters. The events have also impacted 337 million people. Despite the urgent need for climate finance to support mitigation and adaptation. Estimates indicate that Africa will need \$2.8–\$3 trillion to address climate change by 2030, yet current financing levels remain alarmingly low, with the continent receiving only 3% of global climate finance.
- o **Climate shocks in Eastern Africa:** over the past 10–15 years, Eastern Africa has faced a continuous cycle of climate disasters. Severe droughts began around 2009–2010, leading to widespread food insecurity. This was followed by another drought, extensive flooding in 2019, and, starting in 2021, one of the longest droughts in 40 years, lasting five consecutive seasons. Recently, extreme weather has become so erratic that some regions experience droughts while others face floods at the same time, marking Africa's "new normal." Data since the 1970s shows an increasing trend in climate-related disasters, highlighting the region's escalating climate risks.
- o **Disaster Risk Management (Climatic and non-climatic):** is the application of disaster risk reduction policies and strategies to prevent new disaster risks, reduce existing ones and manage residual risks, contributing to the strengthening of resilience and reduction of disaster losses.
- o **Building resilience:** in Africa, the high costs of disaster response make resilience-building essential. Disasters often lead to chaotic and costly responses, and recurring events like floods followed by droughts hinder recovery efforts. This cycle makes response up to 40% more expensive than investing in prevention. Proactive approaches, such as anticipatory action—distributing resources before droughts or relocating livestock to prevent loss—can strengthen resilience. Social protection measures, seen in other regions, could also help African communities better withstand these repeated climate challenges by integrating them into disaster risk management plans.

A survey on natural hazard-induced disasters impacting the agriculture sector yielded the following results:



A survey on what man-made disasters affect the agriculture sector yielded the following result



Discussion points

- o **Management aspect of disaster:** the management aspect focuses on three main areas: (i) policies and strategies, (ii) coordination, and (iii) institutional alignment. Disaster risk management entities, while users of climate services, primarily serve a coordinating function. They rarely run models themselves, such as hydrological or climate models and add value to the outputs generated by these climate services. Institutional efforts focus on aligning global goals with national priorities, such as creating frameworks that integrate disaster risk reduction with climate change initiatives.
- o **Residual risks:** refer to risks that remain even after mitigation efforts. In global frameworks, the primary responsibility for managing these risks lies with national governments, which are considered the main institutions accountable for handling residual risks. However, IGAD also play a role in managing residual risks, guided by the regional mandates entrusted.
- o **Indigenous knowledge,** particularly practices rooted in African traditions, is actively considered and incorporated into disaster risk management. Rural communities have developed unique methods for managing and averting risks based on generations of



lived experience, making their insights invaluable for enhancing preparedness in emergencies. However, working is being done to incorporate indigenous knowledge into scientific explanation of early warning.

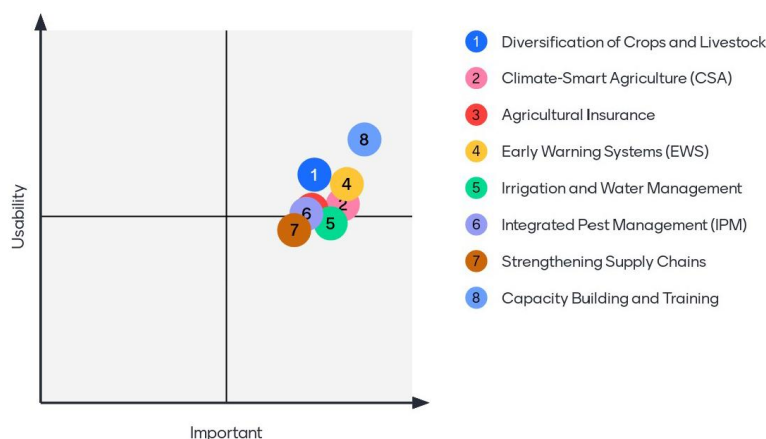
INTRODUCTION TO DISASTER RISK REDUCTION (DRR)

- o **The SENDAI framework:** DRR is guided by global frameworks. In March 2015 the Sendai framework was coined in Japan primarily guiding all DRR interventions. The framework has four priorities for action (i) understanding disaster risks (ii) strengthening disaster risk governance to manage disaster risk (iii) investing in disaster risk reduction for resilience and (iv) enhancing disaster preparedness for effective response and to build back better in recovery, rehabilitation and reconstruction. The framework has seven targets i.e. reduce mortality, affected people, economic loss and damage to critical infrastructure and disruption of basic services and increase countries with national and local DRR strategies, international cooperation and availability and access to multi-hazard early warning systems and disaster risk information and assessments.
- o **Key principles of DRR:** prevention and mitigation efforts focus on reducing risks through measures like improved infrastructure, risk-sensitive land use, and ecosystem-based solutions (NBS). Preparedness is critical to ensure that people and systems can respond effectively to disasters, supported by early warning systems and capacity-building initiatives. Building resilience in individuals, communities, and nations strengthens their ability to withstand and recover from disaster impacts. DRR principles also call for integration within broader development and policy frameworks, incorporating innovative solutions that enhance disaster readiness. Emphasizing DRR as a core development priority, the approach must be community-centered (CBDRM) and scaled to reach all levels, positioning disaster risk reduction as a shared responsibility.
- o **Costs of DRR:** DRR is increasingly challenging and costly, especially as it often involves relocating vulnerable populations. This requires significant investment and well-coordinated systems, including the availability of shelters to safely accommodate people during emergencies. Effective DRR demands meticulous planning, as insufficient preparation can lead to disproportionately high losses even from moderate events, such as category 3 storms, amplifying their impact significantly.
- o **Climate and Disaster Risk Financing (CDRF):** refers to the use of financial instruments and mechanisms designed to provide financial protection against losses due to climate-related disasters. CDRF helps countries, businesses, and individuals manage the financial impact of extreme weather events like droughts, floods, and storms, ensuring they have the resources needed to respond and recover.
- o **DRR funds:** there are a few operational examples worldwide where countries effectively use finance to prepare for, respond to, and recover from disasters. A solidarity fund model is just one approach; another is a seed or subsidy fund, aimed at supporting countries in climate resilience or other innovations that reduce risk. For situations where prevention and mitigation are limited, risk transfer mechanisms—such as crop insurance or index-based livestock insurance—are crucial. Additionally, social



protection mechanisms are common in humanitarian aid, where cash transfers or safety nets are used to support vulnerable communities, helping them sustain their lives and livelihoods. Some countries implement pre-financing mechanisms that set triggers to release funds for response or preparedness, based on pre-arranged criteria.

The participants were asked to rate risk management interventions in terms of importance and usability. The result is shown in the following graph.



Discussion points

- o **Access to finance:** several countries, particularly in the southern region, are struggling to access this due to lack of accreditation. To address this, CCARDESA in collaboration with FAO and SADC have developed a proposal aimed at strengthening the capacities of these countries to achieve accreditation. Additionally, we are exploring opportunities to tap into carbon credit facilities.
- o **Experience on the need for inclusion of collaborators:** in applying for funding from various international organizations, a recurring requirement is the inclusion of collaborators from regions such as Europe, Asia, or the U.S. Incorporating collaborators from these regions as project leads, with local partners in Africa, often simplifies the funding process, as funders are more receptive when their representatives are involved. Attempting to manage projects independently without external collaborators frequently leads to reluctance from funding bodies.

ANTICIPATORY ACTION (AA)

- o **Anticipatory Action** is a systematic way of linking early warning to early action. In AA, climate information is acted upon based on specific thresholds. When a threshold is surpassed, preparation for an impending event becomes necessary. This approach also involves securing pre-financing to support essential preparations. For example, with tropical cyclones, forecasts might indicate a certain level of expected rainfall, allowing for proactive measures to minimize the impact of the extreme event.



- o **Anticipatory action frameworks in Africa:** within the African continent and the IGAD region, various frameworks have been developed to support the anticipatory action and EWS. For example, Within the IGAD region, Sudan, Djibouti, Somalia, Kenya, Uganda and Ethiopia have been reported to have developed their anticipatory action frameworks. Countries in southern Africa such as Zambia, Lesotho, Zimbabwe, Madagascar, Mozambique, Malawi also have frameworks for anticipatory action for different climate change events such as flood, drought, storm etc.
- o **Case studies:** regional case studies have supported WFP in piloting AA initiatives. In Uganda's Karamoja region, the climate team identified key triggers to determine when to initiate responses. Similar projects are underway in Uganda, Kenya, Burundi, and Djibouti. In Kenya, inconsistent response thresholds between agencies, such as the National Disaster Management Authority and the Kenya Meteorological Department, have posed coordination challenges. Harmonizing these standards with Kenya's response roadmap will enable more consistent and unified anticipatory action across agencies like WFP.
- o **IGAD Regional Roadmap for AA:** the IRRAA focuses on six key pillars to guide the AA approach. The pillars are (i) a harmonized approach to develop triggers and thresholds for multiple hazards (ii) a harmonized approach to design anticipatory actions (iii) enhanced access to aa f financing mechanisms (iv) communication and advocacy (v) research, evidence and learning pillar and (vi) coordination and legal framework.
- o **Regional Technical Working Group (TWG) for AA:** has been established, engaging a diverse range of partners across sectors. The group remains open to members from across Africa, including Southern, Central, and Northern regions. Current membership is heavily comprised of government officials, reflecting strong participation from community-level government institutions.
- o **East Africa Dialogue Platform on Anticipatory Humanitarian Action:** to broaden engagement beyond member states, an annual dialogue event has been launched. This event, taking place in early October, offers an opportunity for open engagement across Africa.

Discussion points

- o **Customization of the roadmap for AA:** the approach is for regional bodies to set the overall framework, allowing member states to adapt it to their specific needs. This customization process has also been applied to the main framework: the central level provides a tailored version, and regional entities support national entities in adapting it to fit the national context. For instance, engagement is currently underway to assist four countries in customizing the regional framework to their specific circumstances.



OVERVIEW OF ICPAC'S EAST AFRICA HAZARDS WATCH

- o **Background on East Africa Hazard Watch:** ICPAC decided to develop its own open-source system, allowing for shared code and easy collaboration. This approach means that we can make improvements or add new features at any time without incurring high costs. With in-house staff and on-premises servers, all these updates are manageable internally. Recognizing the need for a system that can monitor multiple hazards across the region, we envisioned and ultimately developed a centralized platform known as East Africa Hazards Watch (Figure 1). This online tool is designed to cover a broad range of thematic hazards.
- o **Thematic areas:** the system is organized into various thematic areas. For example, the climate theme focuses on factors like rainfall, temperature, and climate change. Other thematic areas cover sectors such as agriculture, pests, food security, drought, floods, tropical cyclones, and environmental issues. Based on these themes, we're developing a suite of thematic watches. This approach was chosen in response to requests from member states for a unified analysis system. Embedding everything into a single platform would slow down the central system. Hence, ICPAC's DRM-prepared specialized hazard watches allow faster and more efficient monitoring.

Dr. Jully Ouma from ICPAC demonstrated various watch types for the participants, with an overview shown in Figures 1, 2, and 3.

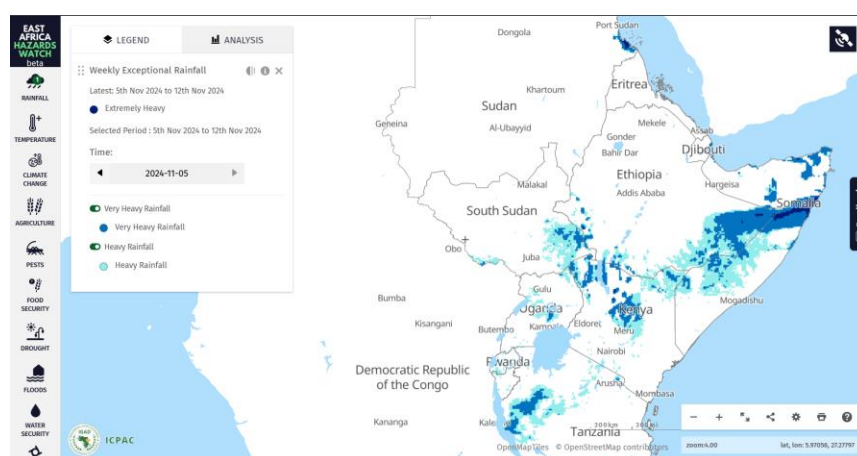


Figure 1. Overview of East Africa Hazards Watch (<https://eahazardswatch.icpac.net/>)

Some thematic area within the East Africa Hazard Watch includes

- o **Rainfall thematic area:** the first key product consistently available is the extreme rainfall forecast, which is produced weekly and has proven valuable to Member States. This forecast identifies exceptionally heavy rainfall and categorizes it into three classes: heavy rainfall, very heavy rainfall, and extremely heavy rainfall. Based on the weekly total rainfall, the product covers a range from 0 up to slightly above 300 mm.



- o **Temperature thematic area:** temperature data is represented as one of the thematic layers, providing the weekly mean temperature for the entire region. This data is updated weekly, delivering an overview of average regional temperatures.
- o **Climate change thematic area:** the climate change section includes three main classes:
 1. Cities temperature evolution - tracks temperature warming trends by city.
 2. Temperature projections - provides projected temperature increases across different climate scenarios and projected average temperatures.
 3. Precipitation projections - shows projected changes in precipitation (as a percentage) and the projected total precipitation, expressed in millimeters per day.
- o **Food security thematic area:** the Integrated Food Security Phase Classification (IPC) Acute Food Insecurity Scale classifies food insecurity found at a specific point in time and of a severity that threatens lives or livelihoods, or both, regardless of the causes, context or duration. This thematic area shows the food security status whenever they release every month, they also make available in the system to showcase where we have food security issues within our region.
- o **Drought thematic area:** identify areas affected by agricultural drought, and areas with the potential to be affected using Combined Drought Indicator.

DISASTER RISK ASSESSMENT

- o **Hazard, exposure, vulnerability, coping capacity, and risk:** A hazard is any event that can cause harm, such as droughts or floods. Exposure is the extent to which people or assets are in the hazard's path. Vulnerability refers to the factors making communities more prone to harm, like poverty. Coping capacity is the ability to manage or recover from hazards, with stronger capacity aiding quicker recovery. Risk combines these elements—the likelihood and severity of harm increase with higher exposure, greater vulnerability, and lower coping capacity.
- o **Climate change vulnerability assessment:** vulnerability is determined by three main factors: exposure, sensitivity, and adaptive capacity. To assess vulnerability to climate change, for example, one approach is to combine data on exposure (such as the number of data points indicating exposure), sensitivity, and adaptive capacity. By integrating these datasets, we can evaluate the overall vulnerability of a community or system to specific hazards or climate change effects.
- o **Risk assessment or risk map** involves integrating layers of data on hazard, exposure, and vulnerability to create a comprehensive understanding of risk in a specific area or scenario. By overlaying these elements, a clear picture of the potential risk is formed. In addition to this method, there are various other approaches to assessing risk. One widely used method involves an equation where risk is calculated as the product of hazard, exposure, and vulnerability, divided by coping capacity.

However, it's important to note that these are few of several possible techniques for assessing risk.

- o **Flood risk profile** is prepared based on a probabilistic forecast of flooding, considering factors such as exposure, hazard, and vulnerability for the IGAD region. As a result, detailed maps have been created. The advantage of this approach is that it also provides precise estimates of the potential impacts including the number of people displaced, flooded cropland (in hectares), affected roads, and the potential effect on GDP, among others.
- o **Flood displacement risk profile** helps to understand displacement of people due to flood. This is prepared with calibrated probabilistic model of flood impacts, particularly on displacement.

A demonstration of the East Africa Drought Watch (Figure 2) was done during the training workshop. The Drought Watch is a near-real time system that uses Earth Observation and Weather Information to monitor drought conditions in the East Africa region.

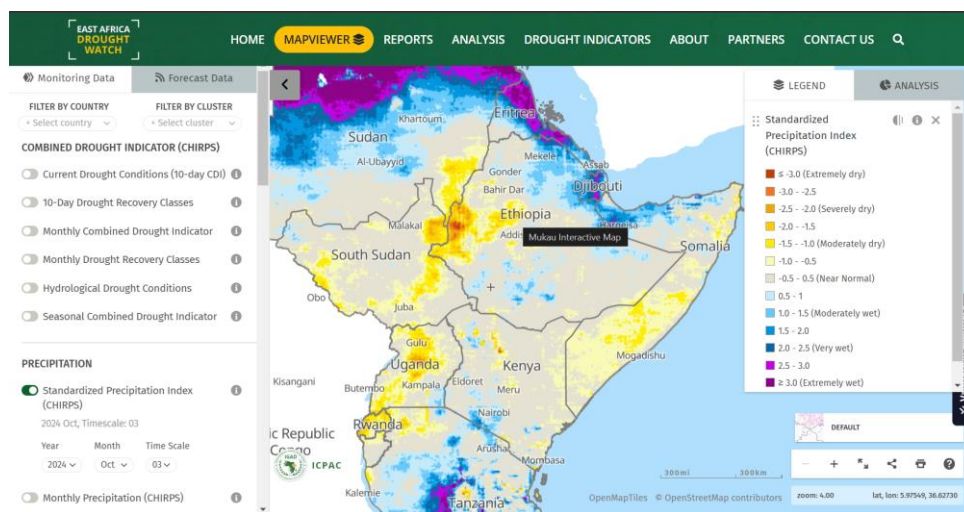


Figure 2. Overview of East Africa Drought Watch (<https://droughtwatch.icpac.net/>)

Drought monitoring is based on the analysis of a series of drought indicators, representing different components of the hydrological cycle (e.g. precipitation, soil moisture, reservoir levels, river flow, groundwater levels) or specific impacts (e.g. vegetation water stress) that are associated with a particular type of drought. The indicators generally represent statistical anomalies of the current situation with respect to the long-term climatology at a given location and period, and so they provide a measure of the probabilistic severity of a given event. The drought indicators monitored are Standardized Precipitation Index (SPI), Soil Moisture Anomaly (SMA), Combined Drought Indicator (CDI) and Anomaly of Vegetation Condition (FAPAR Anomaly).



AGRICULTURE MONITORING FOR EASTERN AFRICA

The East Africa Agriculture Warning Explorer is based on near-real-time Earth Observation and Weather information and provides automatic ten days drought conditions warnings for crops and rangelands at the provincial level. This information and the indicators used are made available as maps and time series statistics in an interactive GIS environment to agricultural and food security analysts. The system provides early warning information on food production anomalies because of climate variability and extremes, particularly relevant with the increasing frequency of climate extremes caused by climate change.

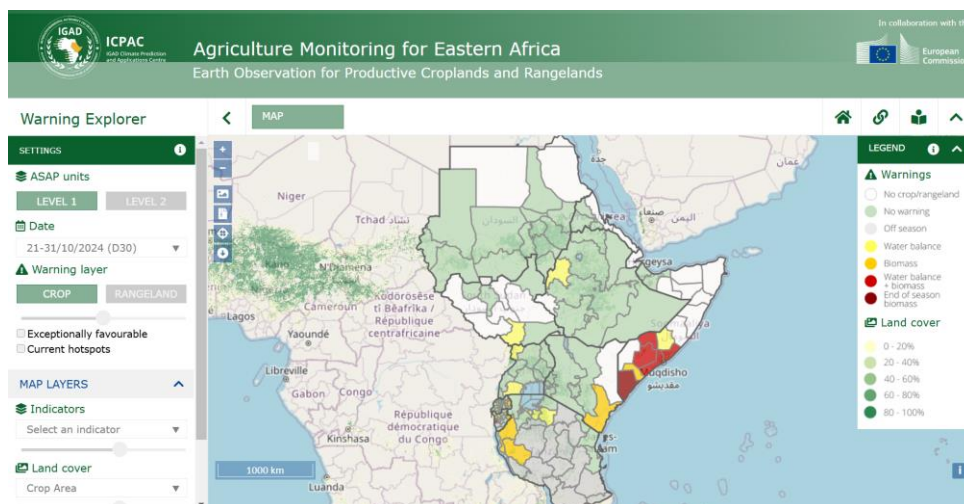


Figure 3. Overview of Agriculture monitoring for Eastern Africa
(<https://agriculturehotspots.icpac.net/>)

Step by step guide to explore the agriculture watch is explained as follows

- o Select the ASAP units level you would like to view. ASAP Level 1 and Level 2 units are ASAP-specific units designed to harmonize the size of the unit of analysis globally. ASAP Level 1 and Level 2 layers are based mainly on the FAO GAUL layers 1 and 2.
- o Select the 10-days period of interest (the most recent period available is the default setting).
- o Select the agricultural land use you are interested in (Crop is the default). Warning levels and statistics are based on the phenologically active area of the selected agricultural land use.
- o Set the opacity level to see the background map behind the warnings.
- o Check the "Exceptionally favourable" box to visualize positive (negative-only is the default) growth conditions.
- o Check the "current hotspots" box to highlight the countries that were marked as hotspots in the last ASAP assessment.
- o Select an indicator to be mapped at the date specified in the field 'Date'.
- o Select the land cover visualized below the warning level (Crop Area is the default). The value is the percentage of each 1 km pixel covered by agricultural area.
- o Select the background map (OSM Mapbox is the default).
- o Click on a region on the map to load statistics related to the ASAP unit.



- o The LEGEND panel shows legends of warnings and indicators
- o Download the map as georeferenced .png (in Web Mercator, EPSG: 3857).
- o If you click on this icon (📍), you activate the indicators query tool: if you click on pixel it will visualize the value of the indicator selected in the Indicators box
- o Download shapefile with the warnings shown in the map
- o The warning class of the unit selected on the map is shown on the left bottom side of the agriculture watch
- o Click on "Explore Stats" to open a new tab where statistics are visualized for the selected unit.
- o Click on Map to go back to the webGIS and select another unit. Click on 'Explore Stats' to open a new tab with the statistics of the new unit. You can switch among tabs to compare different units.
- o Click on 'High Resolution Data' to load and browse high resolution images for the selected unit.

CLIMATE CHANGE FUNDAMENTALS

Dr. Abebe Tadege, principal investigator at ICPAC, presented on the fundamentals of climate change, covering key areas including the mechanisms and causes behind current and projected climate changes, the wide-ranging impacts of climate change, and essential strategies for mitigation and adaptation.

- o **Information and Knowledge requirements for climate change**
debate/negotiations: effective engagement in the climate change debate requires a strong foundation of knowledge and information, including an understanding of the causes of climate change, clear evidence to validate its existence, and reliable long-term climate projections. These projections, which differ from short-term weather forecasts, offer plausible future scenarios that help guide strategic planning. Armed with this knowledge, global and national responses can be crafted, aligning with both local needs and international agreements to address climate challenges comprehensively.
- o **Why should concern be raised about human-caused climate change?** Climate change poses a significant risk to sustainable development and to future generations. The long-term implications are critical. The risks being created today could threaten the very survival of human civilization.
- o **Anthropocene** refers to the impact of humanity on global systems. Major trends such as population growth, economic expansion, and increased transportation are driving significant environmental changes. The era of the Anthropocene is characterized by humans having the capacity to alter the natural environment on a large scale, potentially in ways that are irreversible.
- o **The greenhouse effect:** it refers to the atmosphere's ability to trap heat. There are two types: the natural greenhouse effect, which is essential for life on Earth, and the human-induced greenhouse effect. While the natural greenhouse effect is beneficial, human activities, particularly the release of greenhouse gases from sources like fossil



fuel burning, intensify this effect, creating problems. Due to increased emissions, atmospheric CO₂ levels have now reached 419 parts per million (ppm), which is 50% higher than pre-Industrial Revolution levels. This rise is concerning, especially as trends indicate further increases in the future.

- o **Earth's climate system** is a complex system with five interacting components: the atmosphere (air), the hydrosphere (water), the cryosphere (ice and permafrost), the lithosphere (earth's upper rocky layer) and the biosphere (living things).
- o **Evidence of climate change:** according to Sixth Assessment Report (AR6), GHG concentrations have increased sharply to levels not seen in at least 2 million years. Sea levels are rising at their fastest rate in 3,000 years, Arctic ice has reduced to an extent unseen in over 1,000 years, and glaciers, such as those on Mount Kilimanjaro and other peaks, are rapidly retreating. Another key finding from the IPCC concerns the attribution of climate change. The report shows a clear rise in global temperatures, leading to widespread global warming.
- o **Observed climate trends over the IGAD region:** to understand how climate change is occurring, it's essential to analyze long-term data. Climate change poses a significant risk, and the evidence is clear. Rising temperatures and fluctuating rainfall patterns; some areas experience sharp declines in rainfall, while others see increases is being observed. This is evident in regions like IGAD, where warming and varying precipitation levels are observed.
- o **Climate projections:** Global projections indicate that, regardless of whether we consider the low, medium, or worst-case scenarios, some degree of warming is unavoidable. In terms of rainfall, certain regions are expected to experience increased precipitation, while others are likely to become even drier. According to IPCC projections, global temperatures could rise by up to 5°C by 2100 under a worst-case scenario without intervention or effective policies. In Africa, under the worst-case scenario, significant warming is projected. Eastern Africa, including the Sahara, is expected to see increased rainfall, while Southern Africa is likely to become drier.
- o **Impacts of climate change:** according to IPCC AR6 Working Group II findings the impacts are already more widespread and severe than anticipated, and we are facing an unavoidable increase in these impacts in the near term. As temperatures rise, the risks from climate change will escalate rapidly, often leading to irreversible consequences. Inequities, conflicts, and development challenges further heighten vulnerability to these climate risks. Adaptation is essential, and while feasible solutions exist, greater support is needed to protect the most vulnerable communities. However, some effects of climate change have already reached a point where adaptation is no longer possible. Immediate, decisive action is needed worldwide to address these losses and damages.
- o **Response to Address Climate Change:** the global response to climate change involves a range of policies and strategies adopted by the international community, focusing on mitigation, adaptation, and addressing loss and damage. Key international frameworks include the UNFCCC, the Kyoto Protocol, and the Paris Agreement, all of which aim to coordinate global efforts to reduce emissions and manage climate



impacts. In addition to these international agreements, various climate strategies are also being implemented at regional, national, and local levels, such as NAPs and NDCs, to support tailored approaches to addressing climate challenges.

- o **Adaptation options and response measures for the agricultural sector** include a variety of strategies, such as adjusting planting dates, diversification, agroecology, agroforestry, and climate-smart practices. Additional approaches involve community-based adaptation, improvement of cultivars and species, optimized fertilizer use, insurance, irrigation, and livelihood diversification. Other measures include financial mechanisms, migration support, and strategic relocation, all specifically tailored to enhance resilience in agriculture.
- o **Key negotiation issues at COP29** include reviewing progress on the development and submission of new, ambitious NDCs for the period leading up to 2035. These updated NDCs are required to outline mitigation plans aimed at closing the global emissions gap, targeting a collective reduction of 43% by 2030 and 60% by 2035 (compared to 2019 levels), with the goal of achieving net-zero emissions by 2050. This must be grounded in principles of equity and a just transition away from fossil fuels, alongside ensuring universal access to renewable energy. Additionally, these NDCs must incorporate adaptation plans, strategies for addressing loss and damage, and clearly outline financial provisions to support these goals. In addition, mobilizing international and domestic public and private finance, Global Goal on Adaption (GGA), and operationalizing the Loss and Damage Fund are key negotiation issues at COP29.

Discussion points

- o **Paradox of fossil fuels, economic growth, and sustainable development:** while fossil fuel resources remain available, their use must be managed carefully to prevent harm to environment. Sustainable resource management is therefore essential. Targeted reductions in harmful emissions, for example, play a key role in preserving the ozone layer. Africa now stands at a crucial point, aiming for economic growth through sustainable, viable pathways. The continent's abundant natural resources can be leveraged to advance national and regional interests, but this requires a balanced, responsible approach. By prioritizing emission reductions, we take meaningful steps toward environmental preservation and a sustainable future. In addition, it is also beneficial to look into alternative energy sources.
- o **Access to climate finance:** capacity building is essential, including skills for developing strong, bankable proposals. Many lack the technical expertise needed to create effective proposals, making this a key area for improvement. Additionally, national organization is needed to establish systems that can track funding—monitoring where funds are allocated, how they are utilized, and ensuring accountability and transparency throughout the process.



GENDER-RESPONSIVE MITIGATION AND ADAPTATION STRATEGIES

Joyce Jelagat, a gender and climate expert from ICPAC, talked about the gender-responsive mitigation and adaptation strategies highlighting the following core points.

- o **Gendered impacts of climate change:** the impacts are significant and vary by community roles; in some areas, men are responsible for tasks like water collection, while in others, it falls to women. Increased demands for food and water, coupled with relocation pressures, elevate stress levels and expose individuals—particularly women and children—to risks of violence, assault, and education disruption. Boys may leave school to accompany fathers in search of water and pasture in pastoralist communities, while girls may face early marriage due to economic strain. Resource scarcity also fuels community conflicts, domestic tension, and often leads to malnutrition, disproportionately affecting children, pregnant women, and lactating mothers, creating both immediate and long-term psychological and physical challenges.
- o **Climate change is not gender-neutral:** this means it is essential to ensure that the concerns and challenges faced by different gender groups are considered in the design, planning, and implementation of policies, programs, and projects. Additionally, the contributions and perspectives of these vulnerable groups must not be overlooked.
- o **Gender Results Effectiveness Scale (GRES):** This scale helps in designing adaptation and mitigation projects. We can develop projects that are gender-negative, gender-blind, gender-targeted, gender-responsive, or gender transformative. In designing projects, we should aim for gender-responsive outcomes and, ideally, strive for gender-transformative impact to address pre-existing inequalities and vulnerabilities.
- o **Gender-blind:** gender-blind programs or projects are designed without considering gender differences, potentially reinforcing existing inequalities. These projects don't intentionally discriminate but fail to address or adapt to gender-specific needs, applying a one-size-fits-all approach.
- o **Gender-negative:** describes projects or policies that unintentionally worsen gender inequalities. Rather than being neutral or beneficial, these initiatives may reinforce existing disparities, creating unintended negative consequences for certain gender groups.
- o **Gender-targeted:** gender-targeted initiatives aim to include specific gender participation goals, such as ensuring balanced involvement (e.g., 50 men and 50 women or other target ratios). These projects focus on representation or participation among different genders.
- o **Gender-responsive:** gender-responsive projects actively incorporate measures to address the distinct needs and circumstances of various gender groups, including marginalized groups. They aim to reduce gender disparities through deliberate actions.



- o **Gender-transformative:** gender-transformative projects go a step further, aiming to shift underlying cultural norms, power dynamics, and structural inequalities. They address the root causes of gender-based vulnerabilities to promote long-lasting change.
- o **Gender Mainstreaming Policy Marker:** helps to evaluate a project's gender mainstreaming. This marker assesses how well gender considerations are integrated into a project, assigning it a score from 0 to 2:
 - Gender Marker 0: no deliberate steps have been taken to incorporate gender issues. Gender is not a consideration in the project.
 - Gender Marker 1: gender is a significant, but not primary, objective. While gender issues are addressed in some project components, they are not the main focus.
 - Gender Marker 2: gender equality or equity is the project's primary objective. The project's core purpose is gender-focused, and it would not proceed without this emphasis on gender

WRAP-UP AND VOTE OF THANKS

Before the closing remarks, several participants provided feedback on the workshop. The key themes discussed included:

- o In Kenya, a system is being developed to connect research directly to its end users, particularly farmers. Contacts will be made with the Crop Capability Tool for the Agro-Advisory team and ICPAC, aiming to integrate these systems with the ongoing project. A follow-up with them is planned to explore collaborative opportunities for enhancing the system's effectiveness.
- o A representative from Botswana expressed enthusiasm about seeing forecasts for dry spells, a capability not yet adopted in Botswana. Often, dry spells catch farmers off guard, impacting crops at vulnerable growth stages. Access to this information would support planning to mitigate these risks. The representative emphasized the need for tools and capacity development to implement such forecasting, aiming for an integrated approach in enhancing early warning systems. There is hope for SADC CSC to develop a subregional hazard watch like ICPAC's model.
- o The representative from WMO said all discussions in this training workshop point toward climate action. The insights shared here need to be translated into practice, particularly in promoting sustainable agriculture. This involves rethinking the agricultural value chain, with a focus on integrating climate information and services into climate-smart agricultural practices.
- o Community engagement is essential, as a people-centered approach to applying climate strategies ensures relevance and effectiveness. Engaging communities from the beginning fosters inclusivity and a sense of ownership in project development.
- o Strengthening policy frameworks is also crucial, especially in bridging the gap between scientists and policymakers. Discussions emphasized the importance of ensuring that



parliamentarians actively use provided climate data and facts. Finally, effective disaster management requires robust early warning systems; these tools and processes are vital in addressing the challenges discussed.

- o A comment was made on the Climate Information Services tools introduced, such as the ENACTS Map Room and East Africa Hazard Watch. These tools, which provide region-specific information, are both valuable and engaging. It was suggested that, in the future, a collaborative effort could be made at the regional level to develop a more unified and organized tool.
- o A key point raised was the need to bridge the gap in delivering climate information effectively to end users, particularly farmers. Despite producing a wealth of climate data and high-skill forecasts, the challenge remains in ensuring this information reaches those who need it most. Strengthening communication channels with end users is essential; without this connection, the value of the information is significantly diminished. This area requires focused attention to enhance the impact of climate services.
- o A representative from Central Africa expressed appreciation for the work being done in Eastern and Southern Africa, acknowledging it as impressive and commendable. There is interest in exploring collaboration, particularly in building capacity for Central African countries in CIS. Support in these areas would be greatly valued to strengthen climate-related capabilities in the region.
- o A participant highlighted the valuable experience and knowledge gained, emphasizing the importance of applying these skills effectively in their respective fields. The goal is to ensure that the knowledge acquired here has a practical impact. For example, discussions with Botswana University are underway to incorporate this expertise into their curriculum, aiming to equip future generations with advanced climate knowledge.
- o A noted priority is enhancing in situ observations to verify remote sensing data with ground data, which remains limited in much of Africa. Increased investment in these systems and operations is essential.



CHANGES IN KNOWLEDGE, ATTITUDE, SKILLS, PRACTICES, RELATIONS SURVEY RESULT

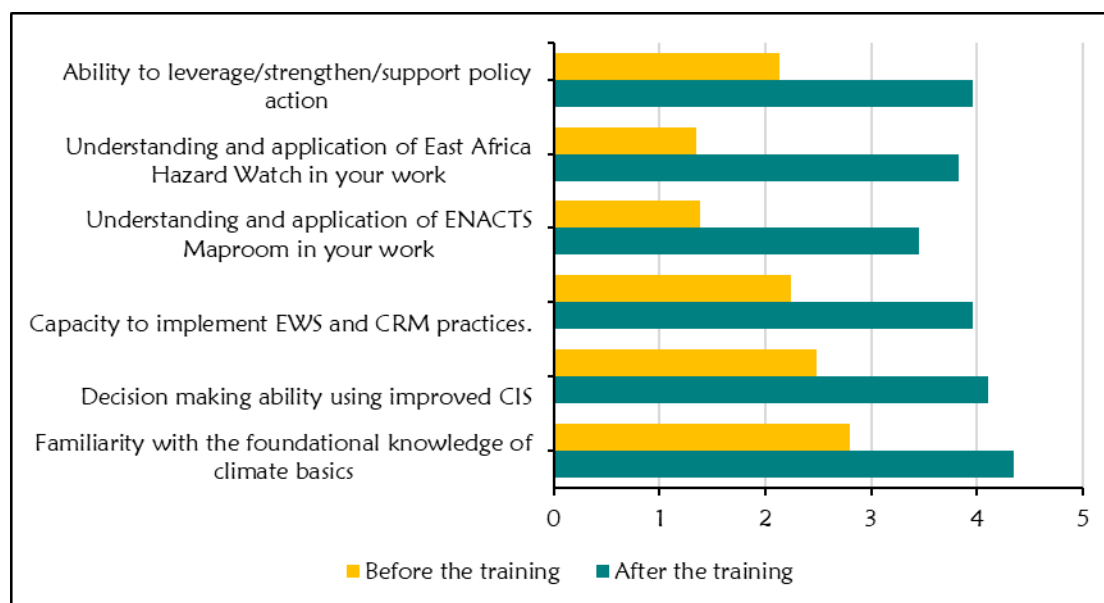


Figure 5. The changes in knowledge, skills, attitudes, and practices on a scale of 1 to 5 (1 = very low, 2 = low, 3 = moderate, 4 = high, 5 = very high)

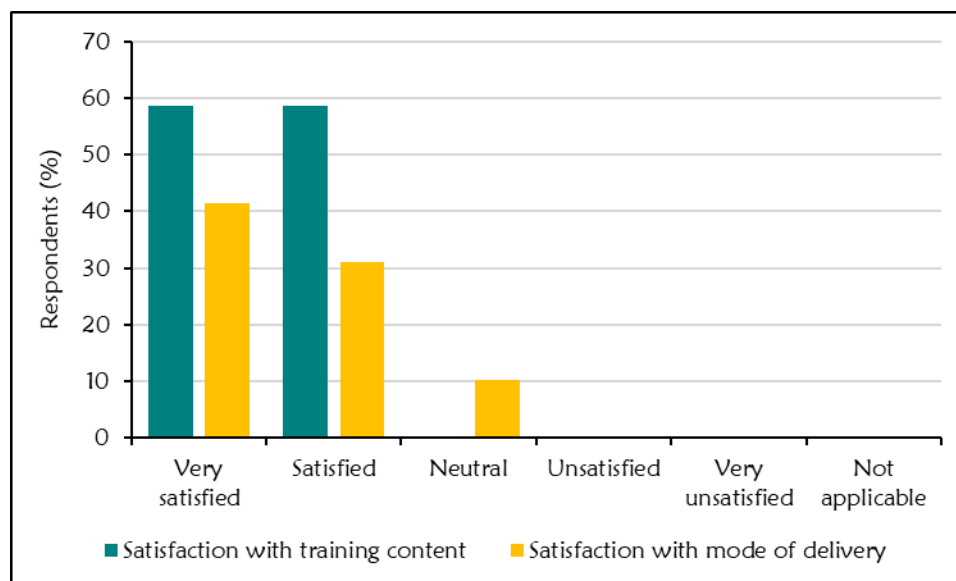


Figure 6. Participants' satisfaction with the training content and mode of delivery

New knowledge, skills, and attributes learned from the training

- o Gained expertise in using and applying climate-related tools such as East Africa Hazards Watch, East Africa Drought Watch, Agriculture Monitoring for Eastern Africa, and



- ENACTS maprooms. Enhanced understanding of their role in disaster risk management, anticipatory action, and disaster risk reduction.
- o Learned to apply tools for modeling climate-related issues and leveraging these tools to address impacts on agriculture, the environment, and other sectors.
 - o Acquired knowledge about various sources and types of climate data and methods for analyzing this data effectively.
 - o Developed skills to navigate and access climate information across multiple platforms.
 - o Understood how to integrate gender considerations into climate information services.
 - o Gained insights into early warning systems and best practices for managing climate risks.
 - o Learned about gender mainstreaming, climate financing strategies, forecasting climate, and the importance of crop suitability maps.
 - o Enhanced understanding of ENACTS maprooms, including the production, interpretation, and application of maps.
 - o Improved proficiency in using climate service tools for accessing and sharing climate information.
 - o Strengthened knowledge of climate systems and their components, using ENACTS maprooms in specific national contexts and forecasting and disaster risk management techniques.
 - o Gained a clearer understanding of the interplay between climate, weather, and agricultural outcomes.
 - o Learned to link digital tools and systems with decision-making processes.
 - o Acquired knowledge about systems developed by ICPAC for forecasting dry spells.
 - o Recognized the importance of involving policymakers in climate forecasting to mitigate the effects of climate change.

Plan for implementing and cascading the knowledge/skills learned from the training

- o Plan to create and implement products that support decision-making in South Africa.
- o Employ crop prediction tools to enhance crop productivity in agricultural practices.
- o Provide early warning information to farmers to help them prepare for climate-related risks.
- o Utilize climate data and tools from the training to incorporate relevant climate information into decision-making.
- o Share the acquired knowledge with students and staff to strengthen community resilience.
- o Organize, record, and analyze climate data and materials for practical use.
- o Train district extension officers and stakeholders on East Africa Hazards Watch, East Africa Drought Watch, Agriculture Monitoring for Eastern Africa, and ENACTS maprooms to enhance implementation efforts.
- o Engage with stakeholders in climate-related fields to foster partnerships.
- o Convene meetings with authorities to advocate for financial support and workshops targeting agricultural officers and climate change programs.
- o Advocate for collaboration with meteorological institutions to improve access to climate data.
- o Support decision-making processes and influence policy development on climate issues.



- o Conduct training on platforms such as Maprooms and East Africa Hazards Watch to empower others.
- o Provide data-driven advice to farmers on mitigation measures to address climate impacts.
- o Apply skills to support regional efforts, including FSRP and other initiatives.
- o Prepare forecasting and early warning reports using tools like Maprooms and East Africa Hazard Watch.
- o Integrate Maproom data into predictions for pest and disease outbreaks.
- o Embed gender considerations into climate change communication and outreach.
- o Organize awareness campaigns on the importance of weather data for informed decision-making.
- o Plan and organize dialogues with high-level policymakers, especially in the Ministry of Agriculture, to address climate challenges and encourage necessary policy actions.

Gaps and suggestions for improvement

Future training opportunities could be enhanced by extending the duration of the sessions to allow for deeper engagement with the material and incorporating more hands-on training, simulations, and practical exercises using real data. Providing participants with training materials in advance and including country-specific case studies would improve preparation and applicability. Adding sessions on creating and applying tools like maprooms for forecasting and focusing on early warning systems and rainfall distribution models would strengthen technical skills. The training should include excursions and increase practical sessions to balance theoretical and applied learning. Encouraging knowledge-sharing through success stories, panel discussions, and group activities would foster collaboration and innovation. Additionally, increasing representation from Central Africa and providing follow-up training sessions would ensure inclusivity and sustained capacity building.

Practical exercises should include creating drought and flood risk profiles, producing crop suitability maps, and exploring nutrient management guided by weather forecasts. Refresher courses on using applications and tools and benchmarking exercises would reinforce learning. Additionally, visiting host country institutions working on climate services would provide real-world exposure.

Special attention is needed to build capacity in SADC CSC for forecasting dry spell duration and intensity and provide Botswana's Ministry of Agriculture with training on modeling for crop selection. Training on improving forecast accuracy and disaster risk management through practical exercises should also be prioritized.

CLOSING REMARK

Moses Odeke, who delivered the closing remark on behalf of ASARECA, noted that this day marks a significant milestone in the shared journey to strengthen climate resilience, expressing hope for the transformative impact. He thanked each participant for their active engagement and commitment throughout the training workshop. Odeke posed a critical question to the group: How can these valuable discussions continue beyond the confines of this room? He suggested the creation of a community of practice within the field, establishing a platform for ongoing sharing, learning, and collaboration.



Odeke expressed special thanks to the facilitators, whose expert presentations and hands-on exercises enabled participants to address the complexities of climate change better. He emphasized that transforming this knowledge into actionable plans for each participant's respective country and organization is the next challenge. He encouraged everyone to apply the insights gained, share experiences, and build collaborations across borders and sectors. Acknowledging that climate change is a shared challenge too great to address alone, Odeke highlighted the need for collaboration, inclusivity, and gender sensitivity in refining strategies for disseminating climate information and services. Odeke extended a special vote of thanks to AICCRA, CCARDESA, ICPAC, ASARECA, SADC CSC, WMO and other partner organizations for organizing the event, and to the Government of Botswana for their generous hosting. In conclusion, Odeke encouraged everyone to leverage their collective expertise to build a more climate-resilient future, also offering thanks for safe journeys to the meeting. He wished everyone a safe return home and continued success in their climate initiatives.

Dr. Abebe Tadege, ICPAC representative, delivered the closing remarks, expressing his excitement about being part of the five-day training workshop. He highlighted the impressive gathering of expertise, bringing together participants from different countries and experts from AICCRA, CCARDESA, ASARECA, ICPAC, WMO and the SADC CSC. This diverse mix of stakeholders reflects the shared responsibility in ensuring food security for Southern and Eastern Africa. He emphasized that the group is tasked with creating the knowledge, tools, and strategies necessary to secure the region's food future. Without success in this endeavor, the region faces significant challenges. The past five days have proven that Africa has immense capacity. The key lies in collaboration and in creating platforms like this training workshop to bring together and harness that potential. He further suggested strong communication is essential for exchanging ideas, sharing tools, and even proposing collaborative projects—steps that can truly make a meaningful impact.

In closing, Dr. Abebe expressed gratitude to all the partners involved for their collaboration in making the training workshop a success. Looking ahead, he proposed that the group reconvene within the next year, perhaps even expanding it, and encouraged the organizers to consider making this happen.

Dr. Sunshine Gamedze, representative of the SADC-CSC, delivered his closing remarks, emphasizing that the work done in this workshop is part of a larger ongoing process. This training workshop continues the training on Sub-Seasonal Prediction for Eastern and Southern Africa that was held before this. He expressed his appreciation for the contributions made by the participants and acknowledged the challenges that arose during the week. He said there's been a clear demand for the products in Eastern Africa, and the pressure is now on SADC CSC to provide similar outputs. This enthusiasm is a great sign, but it also means we must find ways to meet these expectations and ensure that relevant information is accessible to the countries we serve.

He reaffirmed the importance of continued collaboration and mutual support among the partners and stakeholders, stressing the need to keep up the hard work. He said, "We need to continue pushing forward together so that the progress we've made this week leads to lasting change and benefits for the region."



Dr. Dawit Solomon from AICCRA began his closing remarks by expressing his gratitude to the government of Botswana and the event's partners, particularly CCARDESA, ASARECA and SADC CSC. He highlighted the workshop as a significant step forward, expressing how grateful AICCRA is for the opportunity to collaborate. He pointed out that while many organizations discuss climate issues there is always more to learn, and this training workshop provided an important learning experience for everyone involved. Looking ahead, he acknowledged the potential that had been unlocked during the training workshop and stressed the importance of taking the next steps. He emphasized that while the tools presented during the workshop were valuable, the challenge now is to ensure these resources are accessible to the countries in both the IGAD and SADC regions. The priority will be figuring out how these tools can be effectively shared, adopted, and utilized by the countries, ensuring that they have the support they need to make a real impact.

Dr. Dawit also reflected on the broader challenge of funding for African climate and agriculture projects. He raised the important point that while they continue to rely on external donors, it would be transformative if continental financial institutions like the African Development Bank could take a more active role in supporting these efforts. He emphasized that these are the kinds of projects that need substantial investment, and he reiterated AICCRA's commitment to engaging with these institutions to align their resources with the region's needs. He stressed that the work must continue, focusing on educating policymakers at the ministerial level, to ensure that the efforts made here are taken up at the highest levels of decision-making.

In closing, Dr. Dawit expressed his deep gratitude once again for the contributions of all the participants. He thanked everyone for their active participation over the past few weeks and reiterated how much he had learned from the workshop. He looked forward to continuing the collaboration and making even greater strides together.

Dr. Baitisi Kingsley Podisi, CCARDESA representative, began his closing remarks by extending his gratitude to the various representatives and organizations involved in the event. He thanked the AICCRA, ICPAC, ASARECA, Botswana's National Agricultural Research and Development Institute, and all the distinguished delegates in attendance. On behalf of the CCARDESA Secretary, he appreciated all the partners, participants, and Member States who contributed to making the event possible. Dr. Podisi acknowledged that the workshop could not have happened without the support and input of all partners, expressing deep gratitude for the resources, lessons, and engagement that made the event such a success. He highlighted the importance of the interactions and exchanges during the training workshop.

As the facilitator for regional collaboration, Dr. Podisi emphasized CCARDESA's role in coordinating efforts across various regional research organizations, such as ASARECA, CORAF, FARA, and RUFORUM. He shared that CCARDESA has been working closely with these organizations to address important capacity issues, particularly in engaging policymakers and influencing them through various networks. Dr. Podisi also reflected on the importance of CIS and the progress made in Southern Africa. He noted how weather forecasts have become more informative, not just telling the public about weather patterns, but also providing critical information on potential impacts on crops and communities. He



emphasized the need for better communication with the public, to ensure that people understand the significance of weather warnings and forecasts.

Turning to the broader agricultural sector, Dr. Podisi acknowledged the challenges posed by weather variability and the significant losses in the sector due to inadequate weather forecasting and planning. He shared his concern that some policymakers tend to overlook weather forecasts when making decisions, leading to unnecessary losses. He stressed the importance of incorporating weather information into agricultural planning and called for greater attention to the resilience of crops in relation to changing weather patterns. He also highlighted the need for continued research and evidence-based approaches to building resilience in agricultural systems.

Dr. Podisi expressed CCARDESA's deep appreciation for its partnership with AICCRA and other regional initiatives, noting how these collaborations have strengthened their collective efforts in advancing climate-resilient agriculture. The cross-learning and collaboration facilitated by these partnerships are crucial to enhancing regional synergy and improving climate resilience. Before concluding, Dr. Podisi thanked everyone involved in organizing and hosting the event, particularly his colleagues at CCARDESA who worked behind the scenes to coordinate the program.

In closing, Dr. Podisi officially declared the workshop closed, wishing all attendees safe travels and expressing optimism for future collaboration.



ANNEX

Annex 1. Survey Template

CHANGES IN KNOWLEDGE, ATTITUDE, SKILLS, PRACTICES,
RELATIONS/COLLABORATION SURVEY

Title: Regional Capacity Building Training Workshop on Climate And Climate Information For Agricultural Experts In Southern And Eastern Africa Regions

Venue & Date: Gaborone, Botswana | September 16-20, 2024

Part 1: General information

1. How would you describe your occupation?

☐ NGO / Advocacy

☐ Ministry of Agriculture

☐ Private sector

☐ Other, please specify

☐ Research/Academia

.....

☐ National Meteorological and Hydrological Services (NMHS)

2. Gender of the respondent

☐ Male

☐ Rather not say

☐ Women

3. Age of the respondent

☐ Below 30

☐ 41-50

☐ 31-40

☐ Above 51

Part 2: Capturing changes around knowledge, skills, attitudes, and practices

Please rate the following questions in terms of your confidence and knowledge in the given areas on a scale from 1 to 5. (1 = very low, 2 = low, 3 = moderate, 4= high, 5 = very high)

BEFORE the Training					How confident and knowledgeable are you about	AFTER The Training				
1	2	3	4	5		1	2	3	4	5
					Familiarity with the foundational knowledge on climate basics, including the drivers and impacts of climate change on agriculture					
					Decision making ability using improved climate information services					
					Capacity to implement early warning systems and climate risk management practices.					



					Understanding and application of ENACTS Maproom in your work					
					Understanding and application of East Africa Hazard Watch in your work					
					Ability to leverage/strengthen/support policy action					

4. How satisfied are you with the training content?
- ☐ Very satisfied ☐ Unsatisfied
- ☐ Satisfied ☐ Very unsatisfied
- ☐ Neutral ☐ Not applicable
5. Are you satisfied with the mode of training?
- ☐ Very satisfied ☐ Unsatisfied
- ☐ Satisfied ☐ Very unsatisfied
- ☐ Neutral ☐ Not applicable
6. How relevant did you find the sessions of the training to your work on a scale from 1 to 3?
- ☐ 1 = not relevant at all
- ☐ 2 = somewhat relevant
- ☐ 3 = very relevant
7. Overall, did the training meet your expectations?
- ☐ Yes,
- ☐ No, because ...
- ☐ I don't know
8. What new knowledge, skills and attributes did you learn from the training?
-
-
9. How do you plan to implement the knowledge/skills learned from the training in your work environment?
-
-
10. How confident are you in your ability to use the tools e.g East Africa Hazard Watch and ENACTs maproom after this training?
- ☐ Confident
- ☐ Somewhat confident
- ☐ Not confident



11. What is your plan and strategy to cascade the knowledge/skill gained in this training program into your respective institution?

12. What can be improved for future training opportunities?

13. Please share what follow-up training gaps you see and would be valuable

Annex 2. Training Agenda

Date	Activity/ training topic	Responsible
Day 1: September 16		
08:30 – 09:00	Registration	ILRI-AICCRA
09:00 – 09:15	Welcoming remark	AICCRA
09:15 – 09:30	Opening remark	CCARDESA
09:30 – 10:00	Introduction of participants	All
10:00 – 10:15	Overview of the training workshop	AICCRA and ICPAC
10:15 – 10:45	Group photo/Health break	
10:45 – 12:00	Understanding climate systems • Basic climate concepts • Climate system components • Climate processes and variability • Climate drivers and forcing • Types of climate data and climate analysis Interactive discussion	Dr. Tufa Dinku (IRI) Dr. Abebe Tadege (ICPAC) Dr. Hussen Seid (ICPAC)
13:00 – 14:00	Lunch	
14:00 – 15:30	Maproom presentation and discussions	Dr. Tufa Dinku (IRI)
15:30 – 16:00	Health break	
16:30 – 17:00	General discussion on Day 1 engagement	AICCRA and ICPAC
Day 2: September 17		
09:00 – 10:00	Overview of climate information services and key components of CIS;	Dr. Tufa Dinku (IRI)



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	Climate data, analysis and interpretation; Seasonal climate forecasting (forecasting methods, understanding and interpreting the forecast); maproom demonstration	
10:30 – 11:00	Health Break	
11:00 – 12:00	Gender and Climate Information Services	Joyce Jelagat
12:00 – 13:00	General Discussion	Moderator
13:00 – 14:00	Lunch	
14:00 – 15:00	Socio-economic benefits of climate information services in agriculture and food security	Dr. Masilin Gudoshava
15:00 – 16:00	Implementation of climate services • Co-production processes during the Greater Horn of Africa Climate Outlook Forum (GHACOF) • Integrating co-produced climate services in Sectoral platforms	Dr. Hussen Seid (ICPAC)
16:00 – 16:30	Health Break	
16:30 – 17:00	General Discussion	Moderator
Day 3: September 18		
09:00 – 10:30	Pillar I & II of early warning system • Disaster risk knowledge • Detection, observation, monitoring, analysis and forecasting	Dr. Ernest Afesismama (WMO-RoA)
10:30 – 11:00	Health Break	
11:30 – 13:00	Pillar III & IV of early warning system • Warning dissemination and communication • Preparedness and response capabilities	Dr. Ernest Afesismama (WMO-RoA)
13:00 – 14:00	Lunch	
14:00 – 14:30	Crop capability tool for agro-advisory for CSA in Sub-Saharan Africa	Dr. Bradwell Garanganga (Digitron)
14:30 – 15:30	Experience-sharing session	All participants
15:30 – 16:00	Health break	
16:00 – 17:00	Experience-sharing session	All participants
Day 4: September 19		
09:00 – 10:30	Introduction to disaster risk management (definitions and concepts, types of disasters affecting agriculture)	ICPAC



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	Risk Assessment and Analysis (Risk assessment methods, data collection and analysis);	
10:30 – 11:00	Health Break	
11:00 – 13:00	Disaster Risk Reduction (principles of DRR, preparedness and planning, anticipatory action), Risk financing and insurance for Agriculture; Policy and governance in DRM (Agricultural DRM)	ICPAC
13:00 – 14:00	Lunch	
14:00 – 15:30	Overview of ICPAC's East Africa Hazard Watch	ICPAC
15:30 – 16:00	Health Break	
16:00 – 17:00	Discussion	Moderator
Day 5: September 20		
09:00 – 10:30	Climate change fundamentals: Mechanisms and causes of current and future climate changes; Impacts of Climate Change; Mitigation and Adaptation	Dr. Abebe Tadege and Joyce Jelagat (ICPAC)
10:30 – 11:00	Health Break	
11:00 – 13:00	Gender-responsive mitigation and adaptation strategies Expectations from the upcoming COP29	Dr. Abebe Tadege and Joyce Jelagat (ICPAC)
13:00 – 14:00	Lunch	
14:00 – 15:30	General Discussion	Dr. John Recha (AICCRA)
15:30 – 16:00	Key messages from the training workshop	Dr. Teferi Demissie (AICCRA)
16:00 – 16:30	Closing remarks and the way forward	AICCRA and CCARDESA



Annex 3. List of participants

Names	Gender	Organization	Position	Country	E-mail address
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